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SafeGAT: Uncertainty-Gated and Safety-Constrained LLM Supervision for Graph Attention Reinforcement Learning in Traffic Signal Control --Manuscript Draft--

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Abstract:	Deep reinforcement learning (DRL) has advanced adaptive traffic signal control (TSC) by enabling signal policies to respond to dynamic traffic states. However, value-based multiagent controllers remain vulnerable during deployment when Q-value estimates become ambiguous, observations are corrupted, traffic demand shifts from the training distribution, or selected phases violate operational constraints. Recent LLM-assisted TSC methods attempt to improve decision quality by injecting language-model reasoning into the control loop, but most existing designs apply such intervention uniformly or periodically. This creates unnecessary overrides of confident RL decisions, increases inference latency, and may introduce unstable corrections without guaranteeing physical feasibility. To address these limitations, this paper proposes SafeGAT, an uncertainty-gated and safety-constrained supervisory framework for graph-attention reinforcement learning in TSC. SafeGAT retains a GAT-DQN policy as the primary controller and activates LLM-based refinement only when measurable execution risk is detected. Specifically, decision uncertainty is estimated using the margin between the two highest Q-values, abnormal traffic conditions are identified through structured anomaly indicators, and a budgeted Top- intervention allocator restricts LLM calls to the most critical intersections at each control step. The LLM therefore acts as a conditional advisor rather than a direct controller. All candidate actions, whether produced by the RL policy or refined by the LLM, are subsequently projected onto a feasible action set through a safety layer that enforces legal phase membership, valid phase transitions, and minimum green-time constraints. Experiments on SUMO 4×4 and 7×28 traffic networks demonstrate that SafeGAT improves average travel time, queue length, delay, throughput, disturbance robustness, and modeled safety compliance while substantially reducing unnecessary LLM invocations compared with uniform-intervention designs. The results indicate that risk-bounded LLM supervision, when coupled with graphbased RL and hard safety projection, provides a scalable and deployment-aware direction for trustworthy intelligent traffic signal control.

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The Editor-in-Chief

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Dear Editor,

We are pleased to submit our manuscript entitled "SafeGAT: Uncertainty-Gated and Safety-Constrained LLM Supervision for Graph Attention Reinforcement Learning in Traffic Signal Control" for consideration for publication in *Expert Systems with Applications*.

Traffic signal control (TSC) is a critical large-scale sequential decision-making problem. While deep reinforcement learning (DRL) has shown strong potential for adaptive TSC, value-based multi-agent controllers remain vulnerable during deployment when Q-value estimates become ambiguous, observations are corrupted, or traffic demand shifts from the training distribution. Recent LLM-assisted TSC methods attempt to improve decision quality but introduce uniform or frequent overrides that increase latency and may destabilize the learned policy.

This paper proposes SafeGAT, an uncertainty-gated and safety-constrained supervisory framework for graph-attention reinforcement learning in TSC. The key contributions of our work are:

- A strict three-tier authority hierarchy integrating a GAT-DQN policy, selective LLM-based decision refinement, and a deterministic safety projection layer.
- An uncertainty-gated intervention mechanism using Q-value confidence margins and structured anomaly detection (queue spikes, sensor corruption, spillback, and emergency events).
- A budgeted Top-K allocation strategy that restricts LLM refinement to the most critical intersections, bounding inference overhead to $O(KC_LLM)$ independently of network size.
- A deterministic safety projection layer that enforces legal phase membership, valid phase transitions, and minimum green-time constraints for all executed actions.
- Empirical evaluation on SUMO 4×4 and 7×28 traffic networks demonstrating improvements in average travel time, queue length, delay, throughput, disturbance robustness, and modeled safety compliance while substantially reducing unnecessary LLM invocations.

We believe this work is well-suited for *Expert Systems with Applications* given its focus on intelligent systems design, AI-assisted decision making under uncertainty, and real-world deployment considerations in transportation. The manuscript has not been published elsewhere and is not under consideration by any other journal.

All authors have read and approved the manuscript. There are no conflicts of interest to declare.

We look forward to hearing from you.

Yours sincerely,

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SafeGAT: Uncertainty-Gated and Safety-Constrained LLM Supervision for Graph Attention Reinforcement Learning in Traffic Signal Control

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ABSTRACT

Deep reinforcement learning (DRL) has advanced adaptive traffic signal control (TSC) by enabling signal policies to respond to dynamic traffic states. However, value-based multi-agent controllers remain vulnerable during deployment when Q-value estimates become ambiguous, observations are corrupted, traffic demand shifts from the training distribution, or selected phases violate operational constraints. Recent LLM-assisted TSC methods attempt to improve decision quality by injecting language-model reasoning into the control loop, but most existing designs apply such intervention uniformly or periodically. This creates unnecessary overrides of confident RL decisions, increases inference latency, and may introduce unstable corrections without guaranteeing physical feasibility. To address these limitations, this paper proposes SafeGAT, an uncertainty-gated and safety-constrained supervisory framework for graph-attention reinforcement learning in TSC. SafeGAT retains a GAT-DQN policy as the primary controller and activates LLM-based refinement only when measurable execution risk is detected. Specifically, decision uncertainty is estimated using the margin between the two highest Q-values, abnormal traffic conditions are identified through structured anomaly indicators, and a budgeted Top- K intervention allocator restricts LLM calls to the most critical intersections at each control step. The LLM therefore acts as a conditional advisor rather than a direct controller. All candidate actions, whether produced by the RL policy or refined by the LLM, are subsequently projected onto a feasible action set through a safety layer that enforces legal phase membership, valid phase transitions, and minimum green-time constraints. Experiments on SUMO 4×4 and 7×28 traffic networks demonstrate that SafeGAT improves average travel time, queue length, delay, throughput, disturbance robustness, and modeled safety compliance while substantially reducing unnecessary LLM invocations compared with uniform-intervention designs. The results indicate that risk-bounded LLM supervision, when coupled with graph-based RL and hard safety projection, provides a scalable and deployment-aware direction for trustworthy intelligent traffic signal control.

1. Introduction

Traffic signal control (TSC) is a large-scale sequential decision-making problem in which each signalized intersection must select phase actions under stochastic arrivals, partial observability, fluctuating demand, and strong spatial–temporal coupling across neighboring junctions. A local phase decision can alter upstream queues, downstream spillback, arterial progression, and network-level delay; therefore, effective TSC requires both local responsiveness and coordinated control. Classical signal-control methods, including fixed-time plans, actuated control, and periodically optimized coordination schemes, provide interpretable and operationally reliable baselines Webster (1958); Robertson (1969); Hunt, Robertson, Bretherton and Royle (1982); Lowrie (1990). However, these methods depend heavily on predefined cycle structures, calibrated traffic assumptions, handcrafted thresholds, or rule-based triggers, making them less effective under rapidly changing demand, congestion propagation, and abnormal traffic events Cools, Gershenson and D’Hooghe (2013); Qadri, Gökçe and Öner (2020).

Reinforcement learning (RL) has emerged as a powerful alternative because it can learn adaptive control policies directly from interaction with microscopic or mesoscopic traffic simulators. Deep RL and multi-agent RL methods have shown strong potential for TSC by mapping high-dimensional traffic states to phase-selection actions and optimizing long-term congestion-related objectives such as queue length, waiting time, travel time, delay, and throughput Sutton and Barto (2018); Li, Lv and Wang (2016); Wei, Zheng, Yao and Li (2018); Wei, Chen, Zheng, Wu, Gayah, Xu and Li (2019a); Chen, Wei, Xu, Zheng, Yang, Xiong, Xu and Li (2020); Wei, Xu, Zhang, Zheng, Zang, Chen, Zhang, Zhu, Xu

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and Li (2019b). Graph-based RL methods further improve scalability by representing intersections as nodes and road connections as edges, allowing graph neural networks or graph attention mechanisms to capture spatial dependency among neighboring intersections Veličković, Cucurull, Casanova, Romero, Lio and Bengio (2018); Yu, Yin and Zhu (2018); Li, Yu, Shahabi and Liu (2018); Mei, Lei, Da, Shi and Wei (2023).

However, despite their empirical effectiveness, value-based RL controllers remain fundamentally limited during deployment. A selected action may have the highest estimated Q-value while still being unreliable when the gap between the best and second-best Q-values is small. In such low-margin states, the policy has weak preference among competing phases, making the final decision sensitive to noise, approximation error, and distribution shift. This weakness becomes more severe in realistic traffic environments where the controller may encounter corrupted sensor observations, sudden queue spikes, emergency movements, demand bursts, lane blockage, or rare congestion patterns that were poorly represented during training. Standard RL policies optimize expected return but do not explicitly quantify decision uncertainty, detect abnormal operating conditions, or guarantee that the executed action satisfies domain constraints. In particular, learned controllers do not inherently enforce minimum green-time requirements, legal phase-transition rules, or safety-oriented fallback behavior Zheng, Zang, Xu, Wei, Yu, Gayah, Xu and Li (2019); Zhao, Dong, Cao and Chen (2024). These limitations are critical because traffic control is not merely an optimization problem; it is also a constrained real-time control problem where invalid or unstable phase changes can degrade mobility and violate operational feasibility.

Large language models (LLMs) have recently been explored as reasoning modules for decision support in traffic control. Their ability to process structured descriptions, compare contextual conditions, and generate rule-like explanations makes them attractive for refining uncertain RL decisions. Existing LLM-assisted TSC studies indicate that language-model reasoning can support traffic-state interpretation and signal-action refinement Lai, Xu, Zhang, Liu and Xiong (2025); Pang, Wang, Pun, Chen and Xiong (2026). Nevertheless, existing LLM-assisted TSC approaches commonly use language-model reasoning as a direct controller, a frequent post-processor, or a uniformly invoked refinement module. This design is inefficient and potentially unstable. First, uniform intervention can override confident RL decisions that do not require correction. Second, frequent LLM calls increase inference latency and make real-time deployment difficult at network scale. Third, unconstrained LLM recommendations may produce actions that are syntactically valid but operationally unsafe. Fourth, if LLM intervention is not tied to measurable risk, the system lacks a principled explanation for when and why reasoning is invoked. Thus, the central challenge is not simply how to add an LLM to an RL controller, but how to regulate LLM intervention so that it is selective, risk-aware, computationally bounded, and safety constrained.

To address this challenge, this paper proposes **SafeGAT**, an uncertainty-gated supervisory framework for graph-attention reinforcement learning in traffic signal control. SafeGAT retains a GAT-DQN policy as the primary controller and introduces an execution-time supervisory layer that intervenes only under measurable risk. The RL policy first produces Q-values and a candidate phase action for each intersection. The confidence of this decision is then assessed using the margin between the two highest Q-values, where a small margin indicates ambiguity in the learned value function. In parallel, structured anomaly indicators detect abnormal traffic conditions such as queue spikes, corrupted observations, downstream spillback, or emergency-related events. These uncertainty and anomaly signals are combined into an intervention score, and a budgeted Top- K allocator selects only the most critical intersections for LLM-based refinement. SafeGAT further separates advisory reasoning from executable control through a safety projection layer. Any action proposed by the RL policy or refined by the LLM is projected onto a feasible action set before execution. This layer enforces legal phase membership, valid phase transitions, and minimum green-time constraints independent of the decision source. This separation between policy optimization and feasible execution follows the core principle of safe and constrained reinforcement learning Altman (1999); Achiam, Held, Tamar and Abbeel (2017); Dalal, Dvijotham, Vecerik, Hester, Paduraru and Tassa (2018); Alshiekh, Bloem, Ehlers, Könighofer, Niekum and Topcu (2018). Consequently, the LLM is allowed to recommend, but the safety layer determines what can be executed. This authority structure is central to the proposed framework: the graph-attention RL policy performs primary optimization, the LLM provides selective reasoning under uncertainty or anomaly, and the safety projection layer guarantees compliance with the modeled operational constraints.

The main contributions of this work are as follows:

1. **SafeGAT framework:** We propose SafeGAT, a risk-bounded supervisory framework that integrates graph-attention RL with selective LLM-based decision refinement for traffic signal control. The framework establishes a strict three-tier authority hierarchy in which the RL policy proposes, the LLM advises conditionally, and the safety projection layer determines the final executable action.

2. **Uncertainty-gated intervention mechanism:** We introduce a Q-value confidence margin, computed as the normalized gap between the two highest-valued phase actions, together with a structured anomaly detector that monitors four categories of state-level abnormality: queue spike anomalies detected via exponentially-smoothed z-score thresholds, sensor corruption flags derived from out-of-range occupancy and negative queue values, downstream spillback indicators triggered when queue length exceeds intersection-specific capacity, and emergency vehicle event flags from simulator or external detection modules. LLM calls are activated only when at least one of these signals is elevated.
3. **Budgeted Top-K allocation:** We develop a budgeted Top-K allocation strategy that restricts LLM refinement to the most critical intersections at each control step, thereby reducing unnecessary intervention and bounding inference overhead to $O(KC_{LLM})$ independently of network size.
4. **Safety projection layer:** We incorporate a deterministic safety projection layer that enforces legal phase membership, valid phase transitions, and minimum green-time constraints for all executed actions, regardless of whether they originate from the RL policy or the LLM advisor. We provide a formal feasibility guarantee (Proposition 1) proving that every executed action lies within the modeled safe action set.
5. **Fully specified LLM refinement pipeline:** We specify the complete LLM refinement pipeline, including the structured prompt template (Appendix A), the output schema, the JSON parsing and regex fallback procedure, and the mapping from the LLM's discrete `final_phase` field back to a signal-control action index. This specification makes the LLM component fully verifiable and reproducible.
6. **Empirical evaluation:** We evaluate SafeGAT on SUMO 4×4 and 7×28 traffic networks using efficiency, robustness, intervention, safety, and latency metrics, and compare it against classical, RL-only, graph-based, and LLM-assisted baselines.

The remainder of this paper is organized as follows. Section 2 reviews classical TSC, RL-based TSC, graph-based coordination, LLM-assisted control, explainable RL, and safe RL. Section 3 formulates TSC as a constrained multi-agent decision problem. Section 4 presents the SafeGAT framework, including graph-attention control, uncertainty estimation, anomaly detection, Top-K LLM intervention, and safety projection. Section 6 describes the experimental setup, baselines, metrics, and implementation details. Section 7 reports quantitative results, ablation studies, robustness analysis, spatial intervention behavior, LLM backend comparison, and latency feasibility. Section 8 concludes the paper and discusses limitations and future directions. Appendix A provides the complete LLM prompt template.

2. Related Work

Traffic signal control has progressed from analytical signal-timing models to adaptive, learning-based, graph-based, explainable, and increasingly reasoning-assisted control systems. Classical methods established the operational foundation of TSC by optimizing signal timing under simplified traffic assumptions. The seminal work of Webster (1958) derived closed-form estimates for cycle length and green split under isolated intersection settings. Fixed-time control was later formulated as a mathematical optimization problem Serafini and Ukovich (1989), while network-level coordination was addressed by TRANSYT and combined signal-control–route-choice models Robertson (1969); Gartner and Al-Malik (1996). Adaptive urban control systems such as SCOOT and SCATS introduced online adjustment of signal plans using detector feedback Hunt et al. (1982); Lowrie (1990). Actuated and decentralized methods, including fuzzy traffic control, swarm-intelligence-based control, and self-organizing traffic lights, improved responsiveness by switching phases according to rule-based or heuristic local triggers Askerzade and Mahmood (2010); Teodorović (2008); Cools et al. (2013). Broader reviews of computational-intelligence-based traffic signal control and modern TSC methods show that these systems remain valuable because of their interpretability and operational maturity, but they typically rely on fixed structural assumptions, handcrafted thresholds, or periodically updated plans Zhao, Dai and Zhang (2011); Qadri et al. (2020). As a result, they are less effective under rapidly changing demand, spillback propagation, large-scale coordination requirements, and abnormal traffic events that require network-aware adaptation rather than isolated local adjustment.

The formulation of TSC as a Markov Decision Process enabled reinforcement learning methods to optimize signal policies through environment interaction. RL provides a general mathematical framework for sequential decision-making with delayed rewards Sutton and Barto (2018), and Q-learning established value-based learning for delayed-reward control problems Watkins (1989). Early tabular and approximate RL approaches for TSC showed that adaptive traffic signal policies can be learned from traffic-state feedback, but scalability was limited by state-space growth Liao

and Cheng (2009). Deep RL allowed high-dimensional traffic observations to be mapped directly to control actions. Deep policy-gradient and value-function methods were explored for traffic light control Mousavi, Schukat and Howley (2017), while neural signal-timing methods showed that deep RL can learn adaptive control from simulated traffic interactions Li et al. (2016); Genders and Razavi (2016); Vidali, Crociani, Vizzari and Bandini (2019); Liang, Du, Wang and Han (2019). DQN-based methods such as IntelliLight demonstrated the feasibility of learning phase-selection policies for traffic control Wei et al. (2018); Zheng, Zang, Xu, Wei, Yu, Kamijo, Yatabe, Du, Xiong, Chen et al. (2018), and stabilization mechanisms such as Double Q-learning, dueling value decomposition, experience replay, and target networks improved value estimation and training stability Van Hasselt, Guez and Silver (2016); Wang, Schaul, Hessel, Hasselt, Lanctot and Freitas (2016); Gao, Shen, Liu, Ito and Shiratori (2017). Diagnostic and survey studies further show that RL-based TSC performance depends strongly on state representation, reward design, demand distribution, and generalization setting Zheng et al. (2019); Zhao et al. (2024).

Multi-agent reinforcement learning (MARL) became a dominant paradigm for large-scale TSC because each intersection can be modeled as an autonomous decision-making agent while still contributing to a shared network-level objective. Early multi-agent systems demonstrated the feasibility of distributed learning for network traffic signal control Arel, Liu, Urbanik and Kohls (2010); El-Tantawy, Abdulhai and Abdelgawad (2013). Coordinated DQN frameworks extended deep value learning to multiple intersections van der Pol and Oliehoek (2016). General MARL algorithms such as MADDPG and MAPPO provide broader foundations for cooperative multi-agent control Lowe, Wu, Tamar, Harb, Abbeel and Mordatch (2017); Yu, Velu, Vinitzky, Gao, Wang, Bayen and Wu (2022). TSC-specific architectures such as PressLight, MPLight, CoLight, and MetaLight incorporate traffic-domain structure into the learning process Wei et al. (2019a); Chen et al. (2020); Wei et al. (2019b); Zang, Yao, Zheng, Xu, Xu and Li (2020). PressLight uses max-pressure-inspired reward design to improve arterial coordination, MPLight improves decentralized large-scale learning, CoLight introduces attention-based communication among intersections, and MetaLight improves transferability across traffic scenarios. Other large-scale MARL studies have investigated decentralized control, cooperative multi-agent learning, recurrent multi-agent policies, and multiple-intersection coordination Chu, Wang, Codecà and Li (2019); Wu, Zhou, Liu, Yuan, Wang and Huang (2020); Liu, Qin, Su, Luo, Wang and Yang (2023). Despite these advances, standard RL and MARL controllers remain primarily reward-driven and reactive. They optimize expected cumulative return but do not explicitly identify low-confidence decisions, detect out-of-distribution traffic states, or guarantee that selected actions satisfy operational constraints such as minimum green time and legal phase transitions during execution.

Graph neural networks provide an effective mechanism for modeling spatial dependency in traffic networks by representing intersections as nodes and roads as edges. Graph Attention Networks (GATs) assign learnable importance weights to neighboring nodes and are therefore suitable for capturing non-uniform influence among adjacent intersections Veličković et al. (2018). In TSC, CoLight demonstrates that attention-based coordination can improve network-level cooperation by allowing each intersection to selectively aggregate information from relevant neighbors Wei et al. (2019b). Spatio-temporal graph models further extend this idea by jointly modeling temporal traffic evolution and spatial propagation, as shown in traffic forecasting architectures such as STGCN, DCRNN, and attention-based spatial-temporal graph convolutional networks Yu et al. (2018); Li et al. (2018); Guo, Lin, Feng, Song and Wan (2019). Recent graph-based and large-scale TSC methods investigate decentralized multi-agent graph reinforcement learning, multi-layer graph deep RL, constrained network partitioning, hierarchical graph attention, and real-time graph-based traffic congestion detection Zhang, Yu, Zhang, Wang, Luan, Guo and Yuen (2023); Wang, Zhu, Zhang, Tian and Zhang (2024); Gu, Wang, Ma, Jia, Mao, Lim and Wong (2024a); Xu and Mao (2025); Zhai, Hao, Cui and Wang (2025); Liu, Su, Wang, Han, Wu, Wang and Qiao (2025); Ye, Wu, Li, Wu, Song and Zhang (2025). Benchmarking frameworks such as LibSignal provide standardized environments and evaluation pipelines for comparing traffic signal controllers across simulators, networks, and datasets Mei et al. (2023); Mei, Lei, Da, Shi and Wei (2024). However, graph attention alone does not solve deployment-time reliability. Attention weights indicate spatial influence, but they do not quantify whether the selected action is uncertain, whether the current state is anomalous, or whether the final executed action is physically feasible. Thus, GNN-based TSC improves coordination but still requires explicit mechanisms for uncertainty-aware intervention and safety-constrained execution.

Recent LLM-assisted control methods attempt to introduce high-level reasoning into traffic signal control by converting traffic states, signal phases, and local congestion information into structured language prompts. LLMLight uses LLM reasoning to interpret traffic conditions and generate signal-control actions from textual state descriptions Lai et al. (2025). iLLM-TSC combines RL and LLMs by allowing an RL controller to propose candidate actions that can be refined using LLM-based reasoning Pang et al. (2026). These approaches indicate that LLMs can provide

semantic decision support and may help in rare or complex traffic scenarios where pure value-function approximation is insufficient. Nevertheless, existing LLM-assisted TSC designs typically lack a strict execution hierarchy. When the LLM is invoked uniformly, periodically, or without a measurable risk trigger, it may override confident RL decisions unnecessarily, increase inference cost, and introduce unstable corrections into the control loop. Moreover, an LLM-generated action is not inherently guaranteed to satisfy traffic-signal feasibility constraints. A syntactically valid phase recommendation may still violate minimum green time, transition legality, or simulator-specific action restrictions. Therefore, LLM reasoning must be regulated by uncertainty-aware gating and separated from final executable control through a safety layer.

Explainability and interpretability are increasingly important for RL-based traffic signal control because learned policies often operate as black-box decision systems. General explainable RL studies emphasize the need to expose why an agent selects an action, how reward structures influence the learned policy, and whether the policy relies on meaningful state features Puiutta and Veith (2020); Heuillet, Boutouth and Bouziri (2021); Milani, Topin, Veloso and Fang (2022); Juozapaitis, Koul, Fern, Erwig and Doshi-Velez (2019). Visualization-based and attribution-based methods have also been used to inspect deep RL behavior and model predictions Greydanus, Koul, Dodge and Fern (2018); Lundberg and Lee (2017). In traffic signal control, interpretable and explainable RL studies attempt to reveal whether the controller responds to meaningful traffic conditions such as queue length, waiting time, pressure, phase state, and neighboring congestion Cai, Fang and Xu (2025); Schreiber, Ramos and Bazzan (2021); Gu, Gao, Zhang, Liu, Yao and Zhou (2024b); Khan, Ghazal, Alyas, Waqas, Raza, Ali, Khan and Abbas (2025). These works are important because they highlight the deployment gap between high numerical performance and operational trust. However, explainability alone does not correct unsafe decisions, regulate external reasoning, or guarantee that selected actions satisfy signal-control constraints. SafeGAT therefore treats interpretability and supervision as related but distinct requirements: uncertainty and anomaly signals decide when additional reasoning is needed, while safety projection determines what can be executed.

Safe reinforcement learning addresses constraint satisfaction in safety-critical decision-making environments. Constrained Markov Decision Processes provide a formal basis for optimizing rewards under cost or safety constraints Altman (1999). Algorithms such as Constrained Policy Optimization (CPO) aim to maintain policy improvement while satisfying expected constraint bounds Achiam et al. (2017). Projection-based safe RL methods correct unsafe continuous actions after policy output Dalal et al. (2018), while shielding techniques use formal safety specifications to override actions that would lead to unsafe states Alshiekh et al. (2018). These approaches are important because they distinguish reward maximization from feasible execution. However, most safe RL methods do not incorporate semantic reasoning modules, do not use graph-attention coordination for large-scale TSC, and do not provide adaptive intervention mechanisms that decide when external reasoning should be invoked. Conversely, LLM-assisted TSC methods provide reasoning but often lack hard safety guarantees. This separation exposes a gap between graph-based coordination, uncertainty-aware intervention, language-model reasoning, and constrained action execution.

SafeGAT is positioned at this intersection. Unlike classical adaptive systems, it learns from interaction rather than relying only on predefined rules Webster (1958); Robertson (1969); Hunt et al. (1982); Lowrie (1990); Zhao et al. (2011); Qadri et al. (2020). Unlike standard RL and graph-attention controllers, it explicitly estimates decision uncertainty and detects abnormal traffic states during execution Sutton and Barto (2018); Watkins (1989); Wei et al. (2018, 2019a); Chen et al. (2020); Wei et al. (2019b). Unlike existing LLM-assisted TSC methods, it does not treat the LLM as a uniform controller or frequent post-processor; instead, LLM refinement is activated only for high-risk intersections selected by a budgeted Top- K gate Lai et al. (2025); Pang et al. (2026). Unlike generic safe RL approaches, it combines safety projection with graph-based multi-agent control and selective LLM reasoning Altman (1999); Achiam et al. (2017); Dalal et al. (2018); Alshiekh et al. (2018). Consequently, SafeGAT addresses scalability, uncertainty, reasoning, and operational safety within a unified supervisory framework in which the RL policy proposes actions, the LLM advises only under measurable risk, and the safety projection layer determines the final feasible action.

3. Problem Formulation

Traffic signal control is formulated as a constrained multi-agent sequential decision-making problem over a road network. The traffic network is represented as a directed graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where each node $i \in \mathcal{V}$ denotes a signalized intersection and each edge $e \in \mathcal{E}$ denotes a road segment or lane group connecting intersections. Let $N = |\mathcal{V}|$ be the number of controlled intersections. Each intersection acts as an agent that selects a signal phase based on local traffic observations and limited neighborhood information. The underlying control process is modeled as a Markov Decision

Process (MDP),

$$\mathcal{M} = (S, A, P, R, \gamma) \quad (1)$$

where S denotes the global traffic state space, $A = \mathcal{A}_1 \times \mathcal{A}_2 \times \dots \times \mathcal{A}_N$ is the joint action space, $P : S \times A \times S \rightarrow [0, 1]$ is the transition probability induced by traffic dynamics and signal actions, $R : S \times A \rightarrow \mathbb{R}$ is the reward function, and $\gamma \in (0, 1]$ is the discount factor. Since each intersection observes only a partial view of the complete traffic network, the practical control setting is partially observable and non-stationary. We therefore adopt decentralized execution with centralized training, where agents execute local phase decisions while training can use network-level experience and shared optimization signals.

At control step t , the local observation of intersection i is defined as

$$s_i^t = [q_i^t, w_i^t, o_i^t, p_i^t, n_i^t] \quad (2)$$

where q_i^t denotes the queue-length vector over incoming lanes or approaches, w_i^t denotes the cumulative waiting-time vector, o_i^t denotes lane occupancy, p_i^t denotes the currently active signal phase, and n_i^t represents neighborhood information aggregated from adjacent intersections. The neighborhood term allows the controller to account for spatial propagation of congestion, spillback, and downstream capacity. The global state is the collection of all local observations, $s^t = \{s_1^t, s_2^t, \dots, s_N^t\}$. Each agent selects a discrete signal phase

$$a_i^t \in \mathcal{A}_i^{legal} \quad (3)$$

where $\mathcal{A}_i^{legal}$ denotes the legal phase set for intersection i , and the joint network action is $a^t = (a_1^t, a_2^t, \dots, a_N^t)$. In practice, not every legal phase is executable at every time step because signal controllers must satisfy operational constraints such as minimum green duration and valid transition logic. Therefore, the executable action set is time-dependent and denoted by $\mathcal{A}_{i,safe}^t \subseteq \mathcal{A}_i^{legal}$.

The local reward is designed to penalize congestion at each intersection. Let $\mathcal{E}_i$ be the set of incoming lanes or approaches associated with intersection i , and let q_e^t be the queue length on lane e at time t . The normalized queue-based reward is defined as

$$r_i^t = -\min\left(\frac{1}{|\mathcal{E}_i|} \sum_{e \in \mathcal{E}_i} \frac{q_e^t}{q_{\max}}, 1\right) \quad (4)$$

where $q_{\max}$ is a normalization constant corresponding to the maximum expected queue length. This reward is bounded in $[-1, 0]$, where values closer to zero indicate lower congestion and better local traffic conditions. The network-level reward is computed as the average reward across all controlled intersections:

$$r^t = \frac{1}{N} \sum_{i=1}^N r_i^t \quad (5)$$

The objective of the RL controller is to learn a policy π_θ that maximizes the expected discounted return,

$$\pi_\theta^* = \arg \max_{\pi_\theta} \mathbb{E}_{\pi_\theta} \left[\sum_{t=0}^{\infty} \gamma^t r^t \right] \quad (6)$$

However, maximizing expected return alone is insufficient for deployment because the policy may select low-confidence or operationally invalid actions under distribution shift, corrupted observations, or rare traffic events. Therefore, the control objective is treated as a safety-constrained execution problem,

$$\max_{\pi_\theta} \mathbb{E}_{\pi_\theta} \left[\sum_{t=0}^{\infty} \gamma^t r^t \right] \quad (7)$$

subject to

$$a_i^t \in \mathcal{A}_{i,safe}^t \quad g_i^t \geq g_{\min} \quad T_i(p_i^t, a_i^t) = 1 \quad \forall i \in \mathcal{V} \forall t \quad (8)$$

where g_i^t denotes the elapsed green time for the current phase, $g_{\min}$ is the minimum green-time requirement, and $T_i(p_i^t, a_i^t)$ is a transition-validity indicator that equals one if the transition from the current phase p_i^t to the selected action a_i^t is legal. These constraints define the feasible action set

$$\mathcal{A}_{i,\text{safe}}^t = \left\{ a \in \mathcal{A}_i^{\text{legal}} \mid g_i^t \geq g_{\min} \ T_i(p_i^t, a) = 1 \right\} \quad (9)$$

SafeGAT further introduces an execution-time intervention constraint. Let $\mathcal{I}_t \subseteq \mathcal{V}$ denote the set of intersections selected for LLM-based refinement at control step t . To ensure that reasoning overhead remains bounded in large-scale networks, the number of LLM-refined intersections is restricted by

$$|\mathcal{I}_t| \leq K \quad (10)$$

where K is the maximum intervention budget per control step. Thus, the problem addressed in this paper is not only to learn a high-performing graph-based RL policy, but also to execute it through a risk-aware supervisory mechanism that selectively invokes LLM refinement under uncertainty or anomaly while guaranteeing that the final action belongs to the modeled safe action set.

4. Proposed Method: SafeGAT

SafeGAT is a risk-bounded supervisory framework for traffic signal control that composes five tightly coupled components: a graph-attention reinforcement-learning backbone, an uncertainty-aware decision gate, a structured anomaly detector, a budget-constrained large-language-model (LLM) advisor, and a safety projection layer that holds unconditional execution authority. The central design principle of SafeGAT is a strict and verifiable *separation of control authority*. The graph-attention RL policy serves as the primary controller and acquires long-horizon signal-control behavior through interaction with the traffic environment. The LLM does not act as a controller; it functions exclusively as a conditional advisor that is activated only when a measurable risk criterion is satisfied. The safety projection layer is the sole entity empowered to commit a phase action to the traffic signal system, doing so only after verifying that the action satisfies all modeled feasibility constraints. This hierarchy prevents uncontrolled LLM intervention, preserves the learned RL policy under normal operating conditions, and guarantees that every executed action lies within the defined safe action set.

Formally, let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ denote the traffic network graph, where each node $i \in \mathcal{V}$ represents a signalized intersection and each edge $(i, j) \in \mathcal{E}$ encodes a road-level spatial dependency. Let s_i^t denote the observed state of intersection i at control step t , $\mathcal{N}(i)$ its neighbor set, and $\mathcal{A}_i^{\text{legal}}$ the set of legally reachable phase actions. At each control step t , SafeGAT computes the final executable phase through the following sequence of constrained mappings, illustrated in Figure 1:

$$Q_i^t = f_\theta(s_i^t, \mathcal{N}(i)) \quad (11)$$

$$a_{\text{RL},i}^t = \arg \max_{a \in \mathcal{A}_i^{\text{legal}}} Q_i^t(a) \quad (12)$$

$$\rho_i^t = \Gamma(Q_i^t, s_i^t, \mathcal{N}(i)) \quad (13)$$

$$\mathcal{I}_t = \text{TopK}_{i \in \mathcal{V}}(\rho_i^t) \quad |\mathcal{I}_t| \leq K \quad (14)$$

$$a_{\text{cand},i}^t = \begin{cases} \Phi_{\text{LLM}}(s_i^t, a_{\text{RL},i}^t, Q_i^t, \rho_i^t) & i \in \mathcal{I}_t \\ a_{\text{RL},i}^t & i \notin \mathcal{I}_t \end{cases} \quad (15)$$

$$a_{\text{final},i}^t = \Pi_{\mathcal{A}_{i,\text{safe}}^t}(a_{\text{cand},i}^t) \quad (16)$$

Here, f_θ is the graph-attention Q-network (Section 4.1), $\Gamma(\cdot)$ is the risk-scoring function that integrates Q-margin uncertainty with structured anomaly indicators (Sections 4.2–4.3), $\Phi_{\text{LLM}}(\cdot)$ is the validated LLM refinement function (Section 4.4), and $\Pi_{\mathcal{A}_{i,\text{safe}}^t}(\cdot)$ is the safety projection operator (Section 4.5). Equations (12)–(16) make the authority structure explicit: the RL policy proposes a candidate action (12); the LLM may refine this proposal only for the at-most K highest-risk intersections identified in (14); and the safety projection operator (16) determines the final executable phase unconditionally. This design converts LLM usage from uniform post-processing into risk-triggered supervisory correction whose computational footprint is bounded independently of network size.

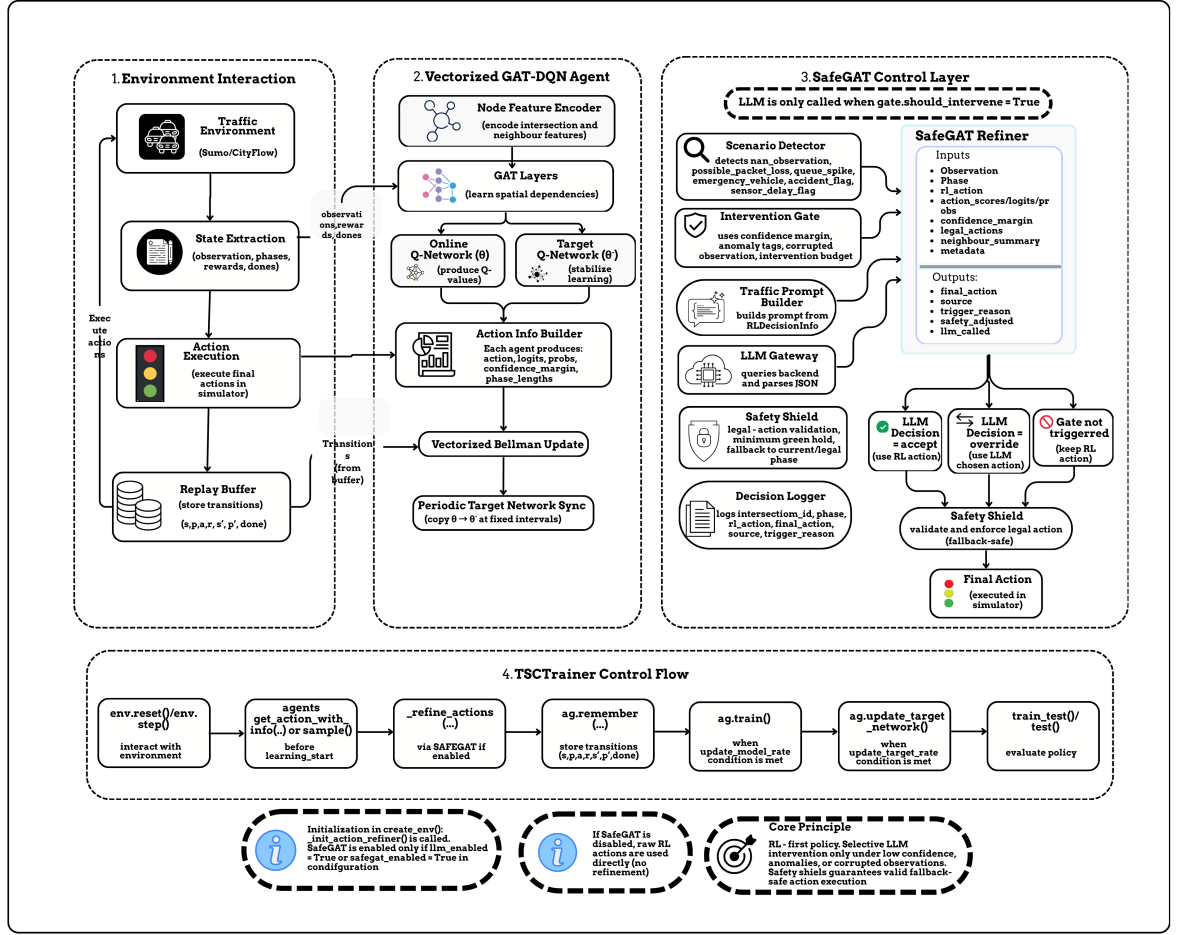


Figure 1: Overall SafeGAT framework. The graph-attention RL backbone produces candidate phase actions and Q-value vectors for every intersection. The uncertainty/anomaly gate evaluates decision reliability and identifies the Top- K highest-risk intersections for LLM refinement. The safety projection layer maps all candidate actions onto the feasible signal-control set before execution, ensuring that neither the RL policy nor the LLM advisor can commit an infeasible action.

4.1. Graph-Attention Reinforcement Learning Backbone

A fundamental challenge in multi-intersection signal control is that congestion propagation is spatially heterogeneous: the influence of a neighboring intersection on a focal node depends on current traffic loading, queue pressure, and network topology, none of which are stationary. Fixed-weight aggregation schemes such as mean pooling or standard graph convolutional networks cannot adapt this influence to changing conditions. SafeGAT therefore adopts graph attention networks (GAT) as the spatial encoder, allowing the model to learn time-varying, neighbor-specific importance weights directly from traffic observations. This adaptive aggregation is especially important near the boundary between uncongested and saturated conditions, where a single overloaded neighbor can dominate downstream queue dynamics.

The traffic network is modeled as $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where each directed edge $(i, j) \in \mathcal{E}$ encodes a road-level spatial dependency between adjacent intersections. The feature vector $x_i^t \in \mathbb{R}^{d_x}$ of intersection i at step t is constructed from local traffic observations comprising per-lane queue length, cumulative vehicle waiting time, lane occupancy, the current active signal phase, and a summary of the immediate neighborhood context. For attention head $m \in \{1, \dots, M\}$,

the unnormalized attention score between intersection i and neighbor $j \in \mathcal{N}(i)$ is computed as

$$e_{ij}^{m,t} = \text{LeakyReLU}\left(\mathbf{a}_m^\top \left[W_m x_i^t \parallel W_m x_j^t \right]\right) \quad (17)$$

where $W_m \in \mathbb{R}^{d_h \times d_x}$ is a learnable feature-projection matrix, $\mathbf{a}_m \in \mathbb{R}^{2d_h}$ is the learnable attention vector for head m , and $\parallel$ denotes concatenation. The score in (17) is then normalized across all neighbors of i via softmax:

$$\alpha_{ij}^{m,t} = \frac{\exp(e_{ij}^{m,t})}{\sum_{k \in \mathcal{N}(i)} \exp(e_{ik}^{m,t})} \quad (18)$$

where $\alpha_{ij}^{m,t}$ reflects the relative contribution of intersection j to the representation of intersection i at time t under head m . The multi-head graph-attention embedding is obtained by concatenating the outputs of all M heads:

$$h_i^t = \parallel_{m=1}^M \sigma \left(\sum_{j \in \mathcal{N}(i)} \alpha_{ij}^{m,t} W_m x_j^t \right) \quad (19)$$

where $\sigma(\cdot)$ is a nonlinear activation and $h_i^t \in \mathbb{R}^{Md_h}$ encodes both local conditions and their spatially weighted neighborhood context. The Q-network maps this embedding to action values as in Eq. (11), and the RL-proposed action is selected greedily from the legal phase set:

$$a_{\text{RL},i}^t = \arg \max_{a \in \mathcal{A}_i^{\text{legal}}} Q_i^t(a) \quad (20)$$

Restricting the argmax to $\mathcal{A}_i^{\text{legal}}$ ensures that the RL proposal is always a reachable phase, even before the safety projection layer in Section 4.5 applies the full feasibility constraints.

The GAT-DQN backbone is trained using experience replay and a periodically updated target network. For a transition tuple (s^t, a^t, r^t, s^{t+1}) sampled from replay buffer $\mathcal{D}$, the temporal-difference target is

$$y_i^t = r_i^t + \gamma \max_{a'} Q_i(s_i^{t+1}, a'; \theta^-) \quad (21)$$

where $\gamma \in (0, 1]$ is the discount factor and θ^- denotes the target-network parameters. Network parameters θ are updated by minimizing the mean-squared Bellman loss:

$$\mathcal{L}(\theta) = \mathbb{E}_{(s,a,r,s') \sim \mathcal{D}} \left[(y_i^t - Q_i(s_i^t, a_i^t; \theta))^2 \right] \quad (22)$$

SafeGAT stores the post-projection executed action $a_{\text{final},i}^t$ in $\mathcal{D}$ rather than the unverified candidate, so that the policy gradient signal remains consistent with the behavior actually deployed in the environment. Storing unverified candidates would create a distributional mismatch between training and execution transitions, destabilizing learning in constrained action spaces.

4.2. Uncertainty Quantification and Structured Anomaly Detection

The RL policy is trained to be greedy with respect to the learned Q-function, but greediness does not imply confidence. In regions of the state space that are rarely visited or where the value landscape is nearly flat, the greedy action may be sensitive to observation noise, function-approximation error, or distribution shift. Separately, even a confident RL decision may be inappropriate when the traffic state itself is anomalous, for example, during a sudden queue spike, a sensor fault, or a spillback event that places operating conditions outside the training distribution. SafeGAT therefore combines two complementary diagnostic signals: a Q-value margin that quantifies decisional uncertainty within the learned value function, and a structured anomaly detector that identifies state-level abnormalities independently of Q-value behavior. Activating LLM refinement only when at least one of these signals is elevated ensures that language-model reasoning is reserved for genuinely difficult or anomalous control situations.

Q-value confidence margin. Let $a_i^{(1)}$ and $a_i^{(2)}$ denote the highest-valued and second-highest-valued phase actions for intersection i at step t . The raw confidence margin is defined as

$$\Delta_i^t = Q_i^t(a_i^{(1)}) - Q_i^t(a_i^{(2)}) \quad (23)$$

and its scale-normalized counterpart is

$$c_i^t = \frac{Q_i^t(a_i^{(1)}) - Q_i^t(a_i^{(2)})}{|Q_i^t(a_i^{(1)})| + \epsilon} \quad (24)$$

where $\epsilon > 0$ ensures numerical stability. Normalizing by the magnitude of the top Q-value removes dependence on the absolute reward scale, which varies with training progress. The low-confidence indicator is then

$$u_i^t = \mathbb{1}(c_i^t < \tau_c) \quad (25)$$

where τ_c is the confidence threshold. Setting $u_i^t = 1$ does not prevent the RL action from being executed; it flags the intersection as a candidate for LLM consultation, subject to the budgeted selection rule in Section 4.3.

Structured anomaly detection. Since Q-margin uncertainty alone is insufficient to capture all failure modes, SafeGAT monitors four categories of state-level anomaly. Each category uses a dedicated detection rule applied to the raw observation stream at every control step.

(i) *Queue-spike anomaly.* A queue spike occurs when the observed queue length at an intersection deviates sharply from its recent historical behavior, indicating an abrupt demand surge, an upstream blockage, or the onset of propagated congestion. SafeGAT detects queue spikes using an exponentially-smoothed z-score. Running mean $\mu_{q,i}$ and standard deviation $\sigma_{q,i}$ are updated online at each step using an exponential moving average with smoothing factor α_{ema} :

$$\mu_{q,i} \leftarrow \alpha_{\text{ema}} \mu_{q,i} + (1 - \alpha_{\text{ema}}) q_i^t \quad \sigma_{q,i} \leftarrow \sqrt{\alpha_{\text{ema}} \sigma_{q,i}^2 + (1 - \alpha_{\text{ema}})(q_i^t - \mu_{q,i})^2} \quad (26)$$

The standardized deviation score and spike indicator are

$$z_{q,i}^t = \frac{q_i^t - \mu_{q,i}}{\sigma_{q,i} + \epsilon} \quad A_{q,i}^t = \mathbb{1}(z_{q,i}^t > \tau_q) \quad (27)$$

where τ_q is a scalar threshold (set to 3.0 in all experiments). A positive spike indicator $A_{q,i}^t = 1$ means that the current queue is more than τ_q standard deviations above the intersection's own running mean, a pattern that the training-time value function has rarely encountered.

(ii) *Sensor-corruption flag.* Sensor faults, communication errors, and loop-detector malfunctions can deliver physically impossible observation values to the controller. SafeGAT checks for invalid measurements by testing range constraints that must hold for any physically realizable traffic state:

$$A_{\text{corrupt},i}^t = \mathbb{1}(o_i^t < 0 \vee o_i^t > 1 \vee q_i^t < 0) \quad (28)$$

where o_i^t is lane occupancy (which must lie in $[0, 1]$) and q_i^t is queue length (which must be non-negative). Any violation of these hard physical bounds flags the observation as corrupted. The indicator $A_{\text{corrupt},i}^t = 1$ signals that the RL policy is receiving unreliable state information and that its value estimates may not reflect true traffic conditions.

(iii) *Spillback indicator.* Downstream queue saturation, commonly called spillback, occurs when a lane or approach fills to capacity and queues extend upstream to block movement at adjacent intersections. Once spillback begins, the network enters a non-linear regime in which locally greedy phase decisions can worsen total delay. SafeGAT detects spillback by comparing the current queue length against an intersection-specific capacity threshold q_{max} :

$$A_{\text{spill},i}^t = \mathbb{1}(q_i^t > q_{\text{max}}) \quad (29)$$

where q_{max} is set to the maximum lane storage capacity for intersection i as reported by the simulator's network model. This threshold is static and does not require online estimation, making it immediately applicable at deployment time without a warm-up period.

(iv) *Emergency-event flag*. Emergency vehicles, accident clearance operations, and other high-priority traffic events require phase decisions that may deviate from the congestion-minimizing policy learned during training. SafeGAT incorporates an emergency indicator $A_{\text{emer},i}^t \in \{0, 1\}$ that is set to one when an emergency or accident flag is raised for intersection i . In simulated environments this flag is read directly from SUMO's vehicle-type and event notification interface; in physical deployments it would be provided by an external incident management system or by a dedicated emergency-vehicle detection module. The flag enables SafeGAT to activate LLM consultation even when Q-margins appear adequate, because the learned value function was not trained on emergency-priority scenarios.

The four indicators are combined into an aggregate anomaly score:

$$A_i^t = A_{q,i}^t + A_{\text{corrupt},i}^t + A_{\text{spill},i}^t + A_{\text{emer},i}^t \quad (30)$$

so that $A_i^t \in \{0, 1, 2, 3, 4\}$, with higher values indicating concurrent anomaly types. Combining u_i^t and A_i^t allows SafeGAT to trigger LLM refinement when the value function is uncertain, when the traffic state is abnormal, or when both conditions hold simultaneously.

4.3. Intervention Scoring and Budgeted Top-K Selection

Invoking the LLM at every intersection at every control step would introduce a reasoning cost that scales linearly with $|\mathcal{V}|$, rendering the framework computationally infeasible for large-scale networks. More fundamentally, uniform invocation would direct LLM reasoning toward intersections that require no intervention, wasting its capacity and potentially destabilizing well-functioning controllers. SafeGAT addresses this by assigning each intersection a scalar intervention priority score and restricting LLM consultation to the at-most K intersections with the highest scores, where K is a fixed budget independent of $|\mathcal{V}|$.

For intersection i at step t , the risk score aggregates the confidence, anomaly, and congestion signals derived in Section 4.2:

$$\rho_i^t = \lambda_u u_i^t + \lambda_a A_i^t + \lambda_q \hat{q}_i^t + \lambda_w \hat{w}_i^t + \lambda_s A_{\text{spill},i}^t \quad (31)$$

where $\lambda_u, \lambda_a, \lambda_q, \lambda_w, \lambda_s \geq 0$ are non-negative weighting coefficients, $\hat{q}_i^t \in [0, 1]$ is the normalized queue length, and $\hat{w}_i^t \in [0, 1]$ is the normalized cumulative waiting time. The score prioritizes intersections that are simultaneously uncertain, anomalous, and congested, reflecting the intuition that LLM refinement is most valuable when multiple risk factors co-occur. The intervention set is then formed as in (14):

$$\mathcal{I}_t = \text{TopK}_{i \in \mathcal{V}}(\rho_i^t) \quad |\mathcal{I}_t| \leq K \quad (32)$$

Two boundary conditions enforce conservatism: any intersection with $\rho_i^t = 0$ is excluded from $\mathcal{I}_t$ even if the budget is not exhausted, preventing LLM invocation when no measurable risk is present; and if fewer than K intersections have positive risk scores, only those are refined. Together, these conditions guarantee that the LLM is never invoked as a uniform post-processor, and that its reasoning overhead per control step is at most $O(KC_{\text{LLM}})$ regardless of $|\mathcal{V}|$.

4.4. LLM Refinement and Output Validation

Graph-attention RL excels at learning compact, generalizable signal-control policies from repeated interaction, but its decisions are conditioned solely on the encoded state representation and the learned value function. In anomalous or rare operating conditions, precisely the situations identified by the gate in Section 4.2, the RL policy may fall back on patterns reinforced under more common conditions that are suboptimal for the present context. A large language model, by contrast, can reason over rich symbolic descriptions of the traffic state and produce context-sensitive phase recommendations without retraining. The role of the LLM in SafeGAT is therefore not to replace the RL controller but to offer a targeted, locally scoped correction when the risk gate identifies that the RL proposal may be unreliable.

Prompt construction. For each selected intersection $i \in \mathcal{I}_t$, SafeGAT constructs a structured prompt encoding six information categories: (i) intersection identity and simulation context; (ii) lane-level traffic state vectors comprising per-lane queue lengths, occupancies, and cumulative waiting times; (iii) neighborhood occupancy summaries from $\mathcal{N}(i)$; (iv) the RL-proposed phase $a_{\text{RL},i}^t$ and the full Q-value vector Q_i^t ; (v) the normalized confidence margin c_i^t from Eq. (24) and the anomaly tag set A_i^t from Eq. (30); and (vi) the scalar intervention severity score ρ_i^t from Eq. (31). The complete prompt template is provided in Appendix A.

LLM backend and scope of request. All experiments in this paper use **Llama-3.1-8B** as the default LLM backend unless otherwise stated (Section 7 also evaluates four additional backends: Llama-3.3-70B, Llama-4-Scout-17B, Qwen3-32B, and GPT-OSS-120B). The LLM is accessed via a local API endpoint; no proprietary cloud calls are required in the default configuration. The scope of the LLM request is deliberately local: the model is not asked to reason about the global network state or to replace the RL controller. Its sole task is to evaluate whether the RL-proposed phase is appropriate given the supplied risk context and to return a binary accept/override decision together with an alternative phase index if an override is recommended.

Output schema and phase-index mapping. The LLM is instructed to return a response conforming to a strict JSON schema containing exactly three fields: `decision` (string, one of "accept" or "override"), `final_phase` (integer, drawn from $\{0, 1, 2, 3\}$), and `reason` (string, a single concise sentence). The `final_phase` field is an integer index that maps directly and unambiguously to a phase action in $\mathcal{A}_i^{legal}$: index k selects the k -th phase in the intersection's legal phase list, ordered identically to the Q-value vector Q_i^t . This ordering is fixed at network initialization and does not change during an episode, so the mapping is bijective and requires no additional lookup at inference time. When the decision is "accept", the `final_phase` field is ignored and the RL-proposed action is used unchanged.

Parsing and fallback. The LLM output is treated as an untrusted advisory recommendation and is never executed directly. Let $\Phi_{LLM}(\cdot)$ denote the composite LLM-plus-validation function:

$$\tilde{a}_i^t = \Phi_{LLM}(s_i^t, a_{RL,i}^t, Q_i^t, c_i^t, A_i^t, \rho_i^t) \quad (33)$$

The output parser first attempts strict JSON decoding of the raw response string. If strict decoding fails, for example because the model prepends a preamble or wraps the object in markdown fences, a regex-based fallback extracts the `decision` and `final_phase` fields using the patterns `"decision"\s*:\s*"accept|override"` and `"final_phase"\s*:\s*([0-3])`. A recommendation is accepted and applied only if all three validation conditions hold: (1) the schema is well-formed (both required fields are present and have the correct types), (2) the decision field belongs to $\{\text{accept}, \text{override}\}$, and (3) the recommended phase index belongs to $\mathcal{A}_i^{legal}$ at the current step. If any condition is violated, or if the LLM call times out (timeout set to 500 ms in all experiments), the RL-proposed action is used without modification. Following the hybrid LLM–RL override paradigm Mazouze et al. (2025), the candidate action before safety projection is therefore

$$a_{cand,i}^t = \begin{cases} \tilde{a}_i^t & i \in \mathcal{I}_t \text{ output valid decision} = \text{override} \\ a_{RL,i}^t & \text{otherwise} \end{cases} \quad (34)$$

All LLM invocations, including invalid outputs, timeout events, fallback activations, override decisions, and the final executed action — are logged to `safegat_decisions.jsonl`. This logging makes the intervention process fully auditable and supports post-hoc analysis of when, where, and why LLM refinement altered the RL proposal Xiong, Deng, Wang, Cao, Li, Yu, Peng, Lin, Smith, Liu, Huang, Ananiadou and Xie (2025).

4.5. Safety Projection and Final Action Execution

Both the RL policy and the LLM advisor can, in principle, propose a phase action that violates traffic signal feasibility constraints: a phase may be physically unreachable from the current phase, the elapsed green time may be too short to satisfy minimum green requirements, or a recommended index may lie outside the legal action set. Neither component has been formally proven to be constraint-safe, and requiring such proofs from a neural Q-network or a language model would be practically intractable. SafeGAT therefore delegates all feasibility enforcement to a deterministic safety projection layer that acts as the final and unconditional execution authority. This separation means that the safety guarantee does not depend on the internal correctness of either the RL network or the LLM; it depends only on the correctness of the projection operator, which is a closed-form rule amenable to formal verification.

For intersection i at step t , the feasible action set is

$$\mathcal{A}_{i,safe}^t = \left\{ a \in \mathcal{A}_i^{legal} \mid g_i^t \geq g_{min} \text{ and } T_i(p_i^t, a) = 1 \right\} \quad (35)$$

where g_i^t is the elapsed green time of the current phase, g_{min} is the minimum green-time requirement, and $T_i(p_i^t, a) \in \{0, 1\}$ is a transition-validity indicator that equals one if and only if phase a is reachable from the current phase p_i^t . The

Table 1

Per-step computational cost of SafeGAT components compared with a uniform LLM post-processing baseline. $N = |\mathcal{V}|$, $E = |\mathcal{E}|$, M = attention heads, d_h = hidden dimension, K = intervention budget, C_{LLM} = cost of one LLM call.

Component	SafeGAT	Uniform LLM baseline
Graph-attention message passing	$O(EMd_h)$	$O(EMd_h)$
Uncertainty/anomaly gate	$O(N)$	$O(N)$
Top- K selection	$O(N \log K)$	—
LLM consultation	$O(KC_{\text{LLM}})$	$O(NC_{\text{LLM}})$
Safety projection	$O(N \mathcal{A})$	$O(N \mathcal{A})$
Total	$O(EMd_h + KC_{\text{LLM}})$	$O(EMd_h + NC_{\text{LLM}})$

final executed phase is obtained by projecting the candidate action onto $\mathcal{A}_{i,\text{safe}}^t$ as in (16):

$$\Pi_{\mathcal{A}_{i,\text{safe}}^t}(a) = \begin{cases} a & a \in \mathcal{A}_{i,\text{safe}}^t \\ p_i^t & g_i^t < g_{\min} \\ \arg \min_{\bar{a} \in \mathcal{A}_{i,\text{safe}}^t} d(a, \bar{a}) & \text{otherwise} \end{cases} \quad (36)$$

where $d(a, \bar{a})$ is a discrete transition cost between the proposed and nearest feasible phase. The three cases carry clear operational interpretations: if the candidate is already feasible it is executed unchanged; if the minimum green-time constraint is violated the current phase is extended by one step; otherwise the nearest feasible phase under the transition metric is selected.

Proposition 1 (Feasibility Guarantee). *Suppose $\mathcal{A}_{i,\text{safe}}^t \neq \emptyset$ for all $i \in \mathcal{V}$ and all $t \geq 0$. Then for any candidate action $a_{\text{cand},i}^t$ produced by the RL policy, the LLM advisor, or any other source, $a_{\text{final},i}^t = \Pi_{\mathcal{A}_{i,\text{safe}}^t}(a_{\text{cand},i}^t) \in \mathcal{A}_{i,\text{safe}}^t$. That is, every executed phase action satisfies legal phase membership, valid phase-transition constraints, and the minimum green-time requirement.*

Proof. The operator $\Pi_{\mathcal{A}_{i,\text{safe}}^t}(a)$ maps a to an element of $\mathcal{A}_{i,\text{safe}}^t$ in all three cases of Eq. (36): the first case returns a itself, which already belongs to $\mathcal{A}_{i,\text{safe}}^t$ by the guard condition; the second returns the current phase p_i^t , which is a valid self-transition when switching is blocked; and the third returns the argmin over the non-empty set $\mathcal{A}_{i,\text{safe}}^t$. Therefore $a_{\text{final},i}^t \in \mathcal{A}_{i,\text{safe}}^t$ in all cases. $\square$

The non-emptiness assumption is satisfied by construction at well-operated intersections, where maintaining the current phase is always a legal and reachable action. Proposition 1 confirms that the safety guarantee of SafeGAT is independent of the RL network's learned parameters and of the LLM's output: no matter what either component proposes, the projection layer enforces feasibility unconditionally. Three boundary conditions merit explicit discussion. If $\mathcal{A}_{i,\text{safe}}^t = \emptyset$ due to a malformed phase-transition graph or an incorrectly configured $g_{\min}$, SafeGAT defaults to the current phase and raises a configuration fault flag. If the LLM consistently times out or returns invalid output, $a_{\text{cand},i}^t$ reverts to $a_{\text{RL},i}^t$ for all intersections, and SafeGAT degrades gracefully to pure RL control without disrupting operation. Setting $K = |\mathcal{V}|$ reduces SafeGAT to full LLM post-processing; in practice, K should be set well below $|\mathcal{V}|$ to preserve the scalability property.

5. Algorithm and Computational Analysis

Table 1 summarizes the per-step computational cost of each SafeGAT component and compares it against a uniform LLM post-processing baseline that queries the language model at every intersection.

The dominant additional cost introduced by SafeGAT relative to a pure RL baseline is $O(KC_{\text{LLM}})$. Because $K \ll N$ in realistic large-scale networks, this cost is bounded independently of network size, in contrast to the $O(NC_{\text{LLM}})$ cost of uniform post-processing. SafeGAT also produces interpretable execution traces at multiple levels of abstraction. The attention coefficients $\alpha_{ij}^{m,t}$ in (18) expose the spatial influence structure and reveal which neighbors drive each

Algorithm 1 SafeGAT Training and Execution Procedure

Require: Traffic graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$; replay buffer $\mathcal{D}$; GAT-DQN parameters θ , target parameters θ^- ; budget K ; confidence threshold τ_c ; minimum green time $g_{\min}$; risk weights $\lambda_u, \lambda_a, \lambda_q, \lambda_w, \lambda_s$

- 1: **for** each episode **do**
- 2: Reset simulator; observe initial network state s^0
- 3: **for** each control step t **do**
- 4: **for** each intersection $i \in \mathcal{V}$ **do**
- 5: Construct x_i^t from local and neighborhood observations
- 6: Compute GAT embedding h_i^t via Eqs. (17)–(19)
- 7: Estimate $Q_i^t = f_\theta(h_i^t)$; select $a_{\text{RL},i}^t = \arg \max_a Q_i^t(a)$ [Eqs. (11)–(20)]
- 8: Compute c_i^t, u_i^t [Eqs. (24)–(25)]
- 9: Update EMA statistics $\mu_{q,i}, \sigma_{q,i}$; evaluate $A_{q,i}^t$ [Eqs. (26)–(27)]
- 10: Check range constraints; set $A_{\text{corrupt},i}^t$ [Eq. (28)]
- 11: Compare queue to $q_{\max}$; set $A_{\text{spill},i}^t$ [Eq. (29)]
- 12: Read event flags; set $A_{\text{emer},i}^t$
- 13: Aggregate A_i^t ; compute risk score ρ_i^t [Eqs. (30),(31)]
- 14: **end for**
- 15: Select $\mathcal{I}_t = \text{TopK}_{i \in \mathcal{V}}(\rho_i^t)$ [Eq. (32)]
- 16: **for** each intersection $i \in \mathcal{V}$ **do**
- 17: **if** $i \in \mathcal{I}_t$ **then**
- 18: Query LLM with structured prompt (Appendix A); parse and validate output
- 19: Set $a_{\text{cand},i}^t$ via Eq. (34)
- 20: **else**
- 21: Set $a_{\text{cand},i}^t = a_{\text{RL},i}^t$
- 22: **end if**
- 23: Project: $a_{\text{final},i}^t = \Pi_{\mathcal{A}_{i,\text{safe}}^t}(a_{\text{cand},i}^t)$ [Eq. (36)]
- 24: **end for**
- 25: Execute $\mathbf{a}_{\text{final}}^t = (a_{\text{final},1}^t, \dots, a_{\text{final},N}^t)$
- 26: Observe r^t, s^{t+1} ; store $(s^t, \mathbf{a}_{\text{final}}^t, r^t, s^{t+1})$ in $\mathcal{D}$
- 27: Sample minibatch; update θ via Eq. (22); periodically sync $\theta^- \leftarrow \theta$
- 28: Log margins, anomaly tags, LLM decisions, fallbacks, safety corrections, and executed actions
- 29: **end for**
- 30: **end for**

intersection's policy decision. The Q-margin c_i^t in (24) quantifies how decisively the value function prefers the selected phase. The anomaly indicators in Eqs. (27)–(30) identify the specific operating-condition categories that triggered intervention, and the risk score ρ_i^t explains why each intersection was or was not selected for LLM consultation. The logged LLM rationale strings provide a natural-language account of every override decision, and safety-projection logs record when and why the final executed action deviated from both the RL proposal and the LLM recommendation. Together, these signals provide a complete, auditable account of every executed phase decision.

6. Experimental Setup

The proposed SafeGAT framework is evaluated in a microscopic traffic simulation setting using the SUMO traffic simulator. Two grid-based traffic networks are considered to examine both moderate-scale and large-scale control behavior: a 4×4 network with 16 signalized intersections and a 7×28 network with 196 signalized intersections. Both networks use traffic demand with an approximate arrival rate of 1800 vehicles/hour/lane. These two scenarios are selected to evaluate whether the proposed uncertainty-gated LLM supervision remains effective when the number of controlled intersections increases substantially. The 4×4 grid represents a compact multi-intersection control problem

where local coordination and short-range congestion propagation dominate, while the 7×28 grid represents a large-scale arterial network where intervention budgeting, latency control, and spatial selectivity become critical. The reinforcement learning backbone is implemented as a graph-attention DQN controller.

Each intersection is treated as an agent, and the traffic network is modeled as a graph in which neighboring intersections exchange spatial information through graph-attention aggregation. At every decision interval, the agent receives local traffic observations and phase information, estimates phase-action values, and proposes a discrete signal phase. Training follows a centralized-training and decentralized-execution setting. During training, agents interact with the environment, store executed transitions in replay memory, update the Q-network after the learning-start threshold, and periodically synchronize the target network. The trainer executes actions every configured action interval, averages rewards over the interval, updates traffic metrics, and stores transitions using the actually executed action after SafeGAT refinement and safety projection. This design ensures consistency between the behavior policy used to collect experience and the actions evaluated during learning. SafeGAT is integrated as an execution-time action refinement module on top of the RL backbone. For each control step, the RL controller first produces the candidate action together with auxiliary decision information such as action scores, action probabilities, confidence margin, current phase, and phase-length information. The SafeGAT refiner then determines whether the RL proposal should be accepted, refined, or repaired. The scenario detector identifies abnormal traffic conditions such as queue spikes (via EMA z-score, $\tau_q = 3.0$), sensor corruption (via out-of-range checks on occupancy and queue values), downstream spillback (via $q_i^t > q_{\max}$), and emergency vehicle or accident event flags from the SUMO event interface. The intervention gate combines low-confidence decisions, anomaly tags, and corrupted-observation indicators into an interpretable intervention score. LLM refinement is triggered only when this score indicates measurable risk, and the Top-K budget limits the number of refined intersections at each control step. This prevents the LLM from being invoked uniformly across all intersections and keeps the additional reasoning cost bounded in large-scale networks.

The LLM refinement pipeline uses a structured prompt (Appendix A) that asks the language model to act as a traffic-signal control verifier rather than as an unconstrained controller. The prompt contains the intersection identifier, current phase, RL-proposed phase, legal phase set, confidence margin, action scores, anomaly tags, local observation summary, and neighbor summary. The LLM must return a valid JSON object containing the decision, final phase index, and technical reason. The parser first attempts strict JSON decoding and then applies a regex fallback if required. If the LLM output is invalid, malformed, timed out, or recommends an illegal action, the system falls back to the original RL action. Before execution, every candidate action passes through the safety shield. The shield verifies legal phase membership and minimum-green constraints; if the proposed action is illegal, it is replaced by the current legal phase or a valid fallback phase. If the minimum-green hold has not been satisfied, the current phase is maintained. Therefore, neither the RL policy nor the LLM can directly execute an invalid phase. Five LLM backends are benchmarked to study the effect of model scale and reasoning behavior on traffic-control performance: Llama-3.1-8B, Llama-3.3-70B, Llama-4-Scout-17B, Qwen3-32B, and GPT-OSS-120B. Each backend receives the same structured decision context and is evaluated under the same simulator, network, traffic demand, RL policy, intervention gate, and safety-shield configuration. This keeps the comparison focused on intervention behavior, override quality, safety-adjustment frequency, and deployment cost rather than on differences in the traffic environment. All baselines are evaluated in the same SUMO networks to ensure a fair comparison. The considered baselines include classical signal control, adaptive control, plain DQN, graph-based RL without LLM refinement, and LLM-assisted methods with less restrictive or uniform intervention. SafeGAT is evaluated against these baselines to isolate the benefit of combining graph-attention RL, uncertainty-aware gating, anomaly-triggered refinement, and safety projection.

6.1. Implementation Details

The trainer uses a configurable number of training episodes, simulation steps, test steps, action interval, replay-buffer size, learning-start threshold, model-update frequency, target-network update frequency, and model-save frequency. At each decision point, the trainer obtains the current phase from each agent and requests an action together with decision metadata. Before the learning-start threshold is reached, exploratory sampled actions are used and the confidence margin is set to zero. After learning begins, the agent returns action information that includes the selected action, logits or action scores when available, action probabilities, confidence margin, and phase-length metadata. SafeGAT refinement is then applied to the batch of agent actions before the environment step is executed. Rewards are accumulated across the action interval and averaged before being stored in the replay buffer and used for metric updates. The SafeGAT implementation consists of five functional components. First, the scenario detector produces anomaly tags and corruption indicators from the observation stream. Second, the intervention gate scores each decision

using low-confidence, anomaly, and corrupted-observation signals. Third, the prompt builder converts the RL decision context into a structured verifier prompt. Fourth, the LLM gateway queries the selected backend and parses JSON-formatted decisions. Fifth, the safety shield validates the proposed phase against legal-action and minimum-green constraints. Each refined decision is recorded by the decision logger, including intersection ID, current phase, RL action, final action, action source, trigger reason, safety-adjustment flag, confidence margin, action scores, anomaly tags, LLM-call status, LLM decision, gate breakdown, and shield reason.

The risk-score coefficients in Eq. (31) are set to $\lambda_u = 2.0$, $\lambda_a = 1.5$, $\lambda_q = 1.0$, $\lambda_w = 0.5$, and $\lambda_s = 1.0$ for all experiments. The highest weight is assigned to the low-confidence indicator u_i^t because Q-margin uncertainty is the most direct signal that the RL policy's greedy choice is unreliable. The anomaly score A_i^t receives the second-highest weight because concurrent anomaly types (queue spike, sensor corruption, spillback, emergency) indicate state-level abnormality beyond what the value function alone captures. The spillback term $A_{\text{spill},i}^t$ receives a dedicated additive coefficient λ_s because downstream queue saturation is both a strong congestion indicator and a precursor to network-wide propagation, warranting independent emphasis beyond its contribution through A_i^t . Normalized queue length $\hat{q}_i^t$ and normalized waiting time $\hat{w}_i^t$ receive lower weights because they are correlated with the anomaly indicators and serve primarily as continuous congestion proxies that break ties between intersections with identical binary risk scores. These coefficients were validated through a grid search over the 4×4 network and kept fixed for the 7×28 experiments to assess out-of-network generalization. These logs are used for intervention analysis, safety analysis, backend comparison, and explainability.

6.2. Evaluation Metrics

Performance is evaluated using both traffic-efficiency metrics and SafeGAT-specific system metrics. The primary traffic metrics are Average Travel Time (ATT), Queue Length (QL), Delay (D), and Throughput (Th). ATT measures the average time required by vehicles to complete their trips and directly reflects network-level mobility. QL measures the average queue accumulation at controlled intersections. Delay captures the time loss experienced by vehicles relative to free-flow movement. Throughput measures the number of vehicles that successfully complete their trips during the evaluation horizon. These metrics jointly evaluate congestion reduction, travel efficiency, and network discharge capacity. System-level SafeGAT metrics are used to evaluate whether the LLM supervision is selective, safe, and deployable. The intervention ratio measures the fraction of intersection decisions selected for LLM refinement. The override rate measures how often the LLM changes the RL-proposed action. The safety-adjustment rate measures how often the safety shield modifies a candidate action before execution. The safety violation rate records violations of the modeled feasibility constraints. The LLM-call count measures computational demand, while latency statistics measure real-time feasibility. For LLM backend comparison, a composite score is computed using three components: Traffic Efficiency (35%), Intervention Control (40%), and 7×28 Reward (25%). The larger weight assigned to intervention control reflects the central objective of SafeGAT: improving traffic performance without allowing excessive or uncontrolled LLM intervention.

The three component weights reflect the distinct priorities of the SafeGAT evaluation. Intervention Control receives the largest weight (40%) because the central claim of SafeGAT is that LLM reasoning should be selective rather than uniform; a backend that improves traffic metrics through indiscriminate overrides would not demonstrate the targeted advisory behavior that the framework is designed to enforce. Traffic Efficiency (35%) measures the primary operational objective, network-level congestion reduction and is weighted nearly as heavily because intervention selectivity is only valuable if traffic performance is maintained or improved. The 7×28 Reward component (25%) captures scalability: a backend that performs well on the smaller grid but degrades on the 196-intersection network is less suitable for deployment in realistic urban settings. The weights were set before running the backend comparison to avoid post-hoc tuning, and the resulting ranking was verified to be robust to ± 5 percentage-point perturbations of each weight.

7. Results and Discussion

This section evaluates SafeGAT from the perspectives of traffic efficiency, learning stability, benchmark-level comparison, disturbance robustness, component contribution, spatial intervention behavior, LLM backend sensitivity, and deployment feasibility. The objective is not only to show improvements in standard traffic signal control metrics, but also to verify whether those improvements are consistent with the design of SafeGAT. Specifically, the evaluation examines whether uncertainty-gated intervention avoids unnecessary LLM calls, whether anomaly-triggered refinement improves behavior under abnormal conditions, whether the safety projection layer prevents infeasible

Table 2

Performance comparison on the SUMO 4×4 and 7×28 grids. ATT = Average Travel Time (s), QL = Queue Length, Th = Throughput (vehicles). Best results per grid are shown in **bold**.

Grid	Method	ATT (s)	QL	Delay (s)	Th (veh)
4×4	Webster (Fixed)	227	179	206	344
	Actuated/Adaptive	162	119	142	419
	Plain DQN	179	134	157	402
	CoLight	151	110	132	432
	LLMLight	155	114	136	428
	iLLM-TSC	160	118	141	421
	GAT-DQN (no LLM)	151	110	132	432
	SafeGAT	145	101	124	441
7×28	Webster (Fixed)	291	231	264	3879
	Actuated/Adaptive	208	154	182	4731
	Plain DQN	229	172	200	4542
	CoLight	193	140	170	4873
	LLMLight	199	146	174	4813
	iLLM-TSC	205	152	180	4754
	GAT-DQN (no LLM)	193	142	170	4873
	SafeGAT	186	130	158	4980

actions, and whether the Top-K budget keeps LLM supervision scalable on larger traffic networks. All experiments are conducted on the SUMO 4×4 and 7×28 grids. The 4×4 grid represents a moderate-scale multi-intersection setting, whereas the 7×28 grid contains 196 signalized intersections and therefore stresses congestion propagation, intervention budgeting, and real-time feasibility.

7.1. Overall Quantitative Performance

Table 2 shows that SafeGAT achieves the strongest reported performance across the comparable traffic-efficiency metrics on both network scales. On the 4×4 grid, SafeGAT reduces average travel time from 227 s under Webster control to 145 s, corresponding to a 36.1% reduction. Compared with the graph-based RL backbone without LLM supervision, ATT decreases from 151 s to 145 s. Although this improvement over GAT-DQN is smaller than the improvement over classical control, it is methodologically important because GAT-DQN is already a strong learning-based baseline. SafeGAT also reduces queue length from 110 to 101 and delay from 132 s to 124 s while increasing throughput from 432 to 441 vehicles, indicating that the reduction in congestion does not come from suppressing vehicle discharge.

On the 7×28 grid, the benefits remain consistent under a substantially larger control problem. SafeGAT reduces ATT from 291 s to 186 s relative to Webster and from 193 s to 186 s relative to GAT-DQN. Queue length decreases from 142 to 130 compared with the RL-only backbone, and delay decreases from 170 s to 158 s. Throughput increases to 4980 vehicles, which is higher than all reported baselines.

The comparison clarifies the role of each method family. Webster and actuated control provide stable but limited adaptation because their decisions are not learned from the evolving network state. Plain DQN improves adaptivity but lacks explicit spatial coordination. CoLight and GAT-DQN introduce graph-based or attention-based coordination, which explains their strong performance among RL baselines. LLMLight and iLLM-TSC introduce language-model reasoning, but their performance remains below SafeGAT because reasoning alone is insufficient without risk-triggered invocation and feasibility enforcement. SafeGAT improves over these methods because graph attention improves spatial representation, uncertainty/anomaly gating controls when reasoning is needed, and safety projection controls what can be executed.

7.2. Training Stability and Convergence

Figure 2 examines whether SafeGAT maintains stable learning and inference behavior. This is necessary because an LLM-assisted controller can become unstable if the LLM frequently overrides the learned policy and causes the

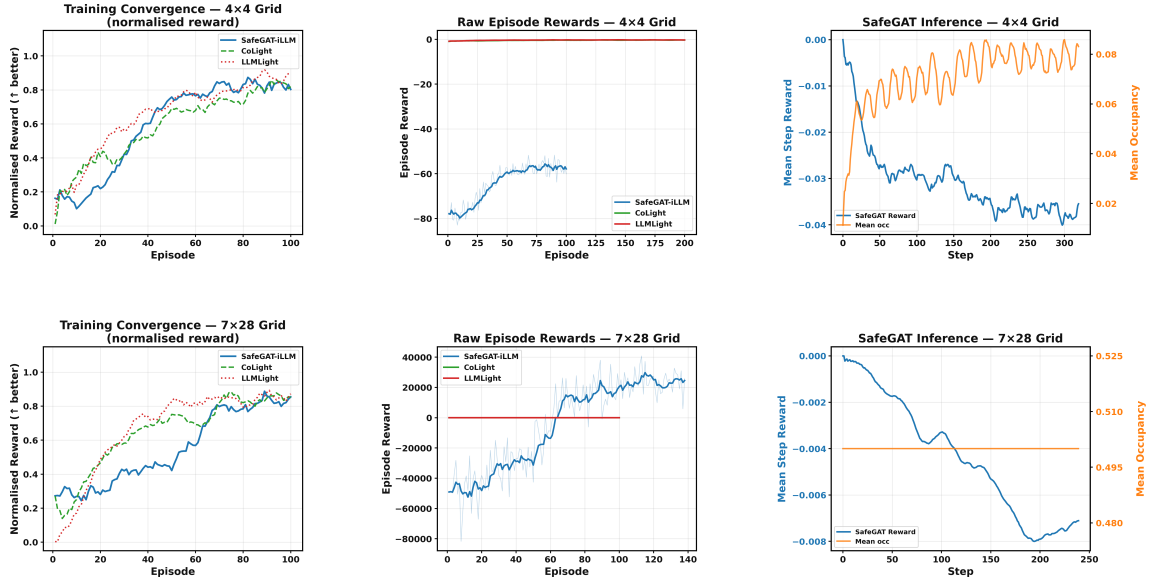


Figure 2: Training convergence and inference stability. Top row: 4x4 grid; bottom row: 7x28 grid. Left: normalised training reward; centre: raw episode rewards; right: SafeGAT inference reward and mean occupancy.

replay buffer to contain actions that are inconsistent with the behavior executed in the simulator. SafeGAT reduces this risk by making the LLM a conditional advisor rather than a continuously active controller. During normal traffic states, the GAT-DQN policy remains responsible for action selection; LLM refinement is activated only when the confidence margin or anomaly detector indicates risk. The replay buffer stores the post-refinement action actually executed in the simulator, reducing mismatch between data collection and value-function updates.

The training curves show that SafeGAT reaches stable reward progression on both grid sizes and remains competitive with CoLight and LLMLight during convergence. The raw reward curves further show stable behavior even in the larger 7x28 grid, where the number of agents and interactions is much higher. The inference plots provide additional evidence: reward behavior remains bounded, and mean occupancy does not diverge after deployment. If the LLM were over-intervening, unstable occupancy oscillations or reward fluctuations would be expected because of frequent phase changes. Instead, the observed stability indicates that the gate prevents unnecessary intervention and that the safety projection layer suppresses infeasible or premature switches.

7.3. Extended Benchmark Comparison

Figure 3 provides a metric-level view of the main comparison. SafeGAT achieves the lowest ATT, lowest queue length, lowest delay, and highest throughput on both grids. This simultaneous improvement is important because TSC methods can show apparent gains in one metric while degrading another. For example, a controller may reduce local queues through aggressive switching while increasing delay due to unstable phase changes. Similarly, a controller may increase throughput on one corridor while creating downstream congestion. SafeGAT avoids these trade-offs because the LLM is not allowed to override every decision and because the safety shield constrains the final executed action.

The improvement over LLMLight and iLLM-TSC is central to the paper's claim. These baselines indicate that LLM reasoning can be useful, but SafeGAT shows that the way reasoning is inserted into the control loop is decisive. Uniform or weakly regulated LLM intervention may modify actions that are already reliable, increasing overhead and destabilizing the learned policy. SafeGAT instead uses the LLM only when the RL policy is uncertain or when anomaly indicators suggest that the current state is outside normal operating conditions. Therefore, the gains in Fig. 3 should be attributed not merely to the presence of an LLM, but to the combination of selective triggering and constrained execution.

7.4. Robustness under Dynamic Disturbances

Figure 4 evaluates SafeGAT under dynamic disturbance injection. This experiment is important because average-case traffic performance alone does not establish deployment reliability. In real traffic networks, controllers must handle

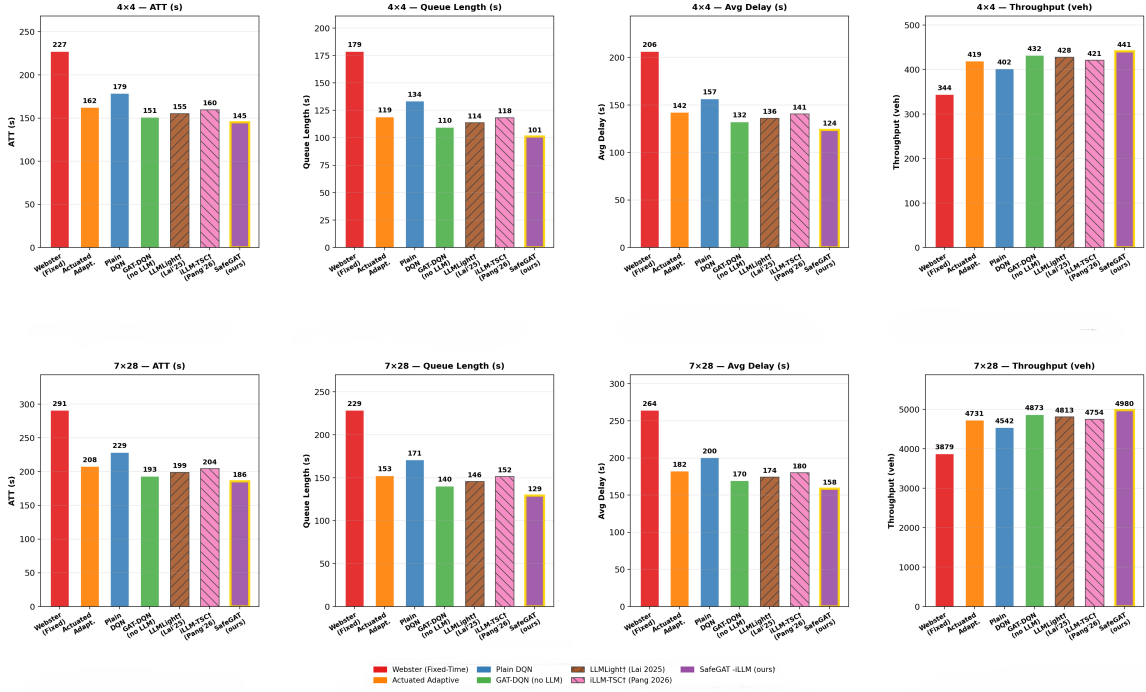


Figure 3: Extended benchmark comparison on 4x4 (top) and 7x28 (bottom) grids across ATT, Queue Length, Avg Delay, and Throughput. SafeGAT achieves the strongest reported result on all four comparable metrics at both scales.

sudden demand bursts, observation corruption, queue spikes, incidents, emergency movements, and other rare events that differ from the training distribution. Pure RL policies usually respond to such events using value estimates learned under previous conditions. If the disturbed state is far from the training distribution, the greedy action may be unreliable even when it has the maximum Q-value.

The disturbance results show that both pure RL and SafeGAT experience degradation after the injection point, but SafeGAT maintains a stronger recovery profile. On the 4x4 grid, the post-injection behavior shows controlled occupancy and increased LLM activity, indicating that the gate identifies the disturbance as a risk condition. On the 7x28 grid, the reward gap between SafeGAT and pure RL increases after the disturbance, showing that the advantage of SafeGAT becomes more pronounced under abnormal traffic dynamics. This behavior is consistent with the intended design: as the environment becomes more uncertain or congested, intervention scores increase, and the Top-K gate allocates LLM refinement to the most critical intersections.

SafeGAT therefore provides an execution-time correction mechanism that does not require retraining the RL policy during the disturbance. The LLM does not learn a new policy; instead, it evaluates high-risk local decisions using structured traffic context. The safety shield prevents the refined action from violating the modeled feasibility constraints. This explains why SafeGAT can respond faster than pure RL value updates under disturbance conditions and supports the claim that uncertainty-gated LLM supervision is most useful in states where learned value estimates are least reliable.

7.5. Ablation Analysis

Figure 5 isolates the effects of LLM intervention, selective gating, and safety projection. The GAT-DQN-only variant (V1) evaluates the graph-attention RL backbone without LLM support. The uniform LLM variant (V2) tests whether simply adding language-model reasoning is sufficient. The full SafeGAT variant (V3) evaluates the complete design with risk-triggered intervention and safety projection. This ablation is central because it separates the effect of the proposed supervisory architecture from the general effect of using an LLM.

The results show that uniform LLM intervention is inefficient and can be harmful. V2 generates substantially more LLM calls than SafeGAT because decisions are refined without checking whether the RL policy is uncertain or whether

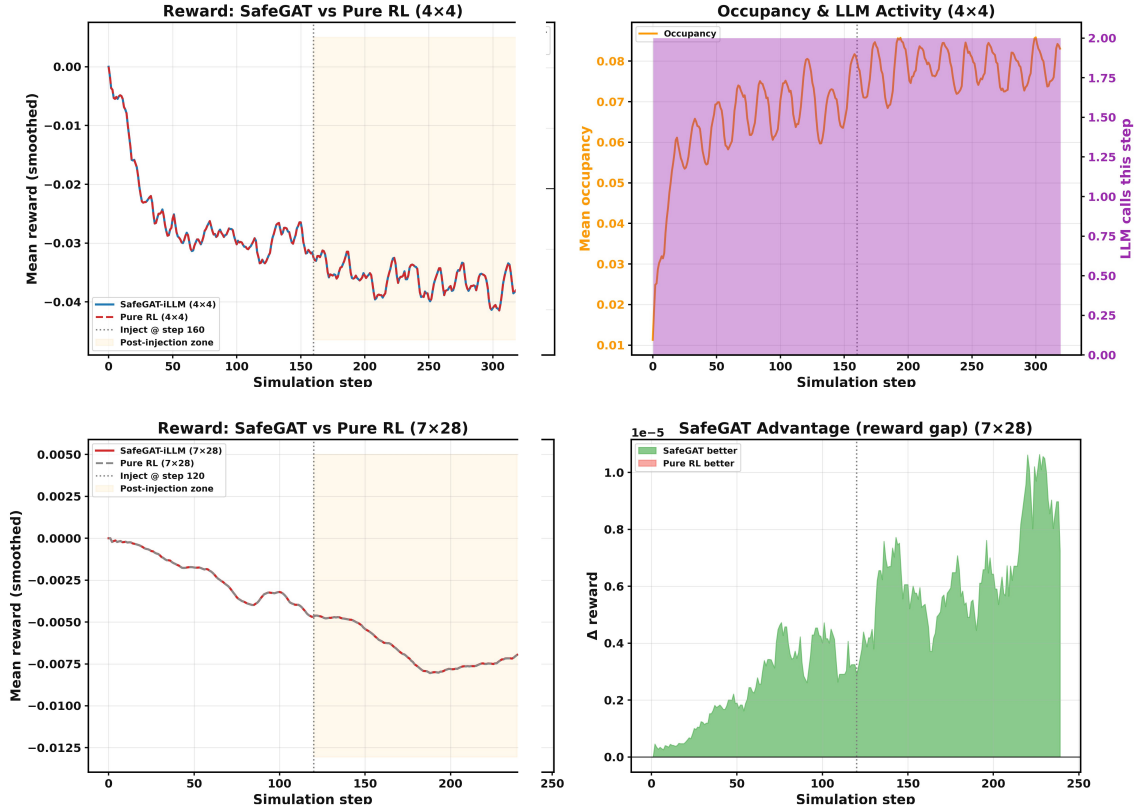


Figure 4: Robustness under disturbance injection. Top (4×4): smoothed reward comparison with injection at step 160; right: occupancy and LLM activity. Bottom (7×28): reward comparison with injection at step 120; right: SafeGAT advantage as reward gap over Pure RL.

the state is abnormal. This increases computational cost and raises the probability of unnecessary overrides, which can disturb stable RL behavior. Full SafeGAT avoids these issues by reducing LLM calls from 2570 to 640 on the 4×4 grid and from 1950 to 490 on the 7×28 grid. This approximately 4× reduction shows that many uniform LLM calls are unnecessary for effective control.

The safety results are equally important. V1 and V2 produce hundreds of safety violations, while V3 eliminates violations under the implemented legal-phase and minimum-green constraints. This shows that neither graph-based RL nor unconstrained LLM reasoning is sufficient for safe execution. The shield is necessary because the final action must satisfy operational feasibility independent of how it was generated. Thus, the ablation validates the main design claim: SafeGAT’s performance does not come from adding an LLM alone; it comes from the interaction of graph-based value learning, selective reasoning, and safety projection.

7.6. Spatial Analysis of Intervention Behaviour

Figures 6 and 7 analyze where SafeGAT intervenes in the network and why those locations are selected. This analysis is necessary because the proposed method claims to be selective rather than uniform. If intervention were evenly distributed across all junctions, the Top- K gate would not be identifying meaningful high-risk regions. The 4×4 results show heterogeneous intervention behaviour: some junctions receive few interventions, while others are corrected more frequently. Intervention rates range from 5.0% (J2) to 42.2% (J6), indicating that the LLM does not act as a blanket controller. The Q-margin heatmap provides stronger evidence for selectivity. The mean Q-margin at LLM invocation is 0.0025, far below the threshold $\tau = 0.05$. This means that the gate is not activating under confident value estimates. Instead, it selects states where the best and second-best phase actions are nearly tied, which are precisely the states where greedy value-based control is fragile. Notably, junctions with higher intervention rates (e.g., J6 at

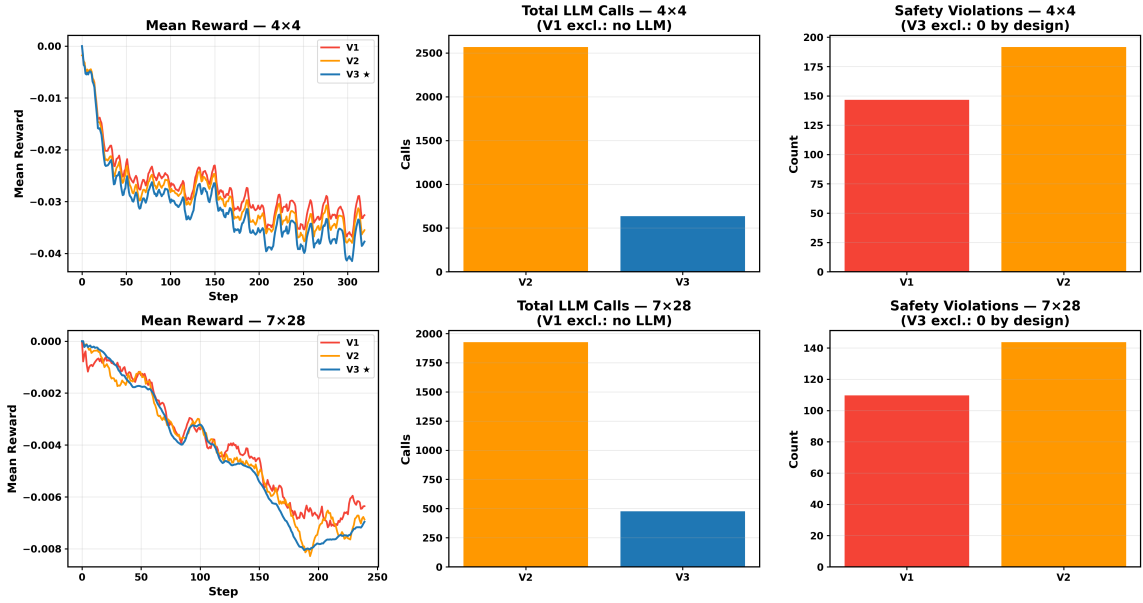


Figure 5: Ablation study on 4×4 (top) and 7×28 (bottom). V1: GAT-DQN only; V2: Uniform LLM without gate or shield; V3: Full SafeGAT. Left: mean step reward; centre: total LLM calls; right: safety violations.

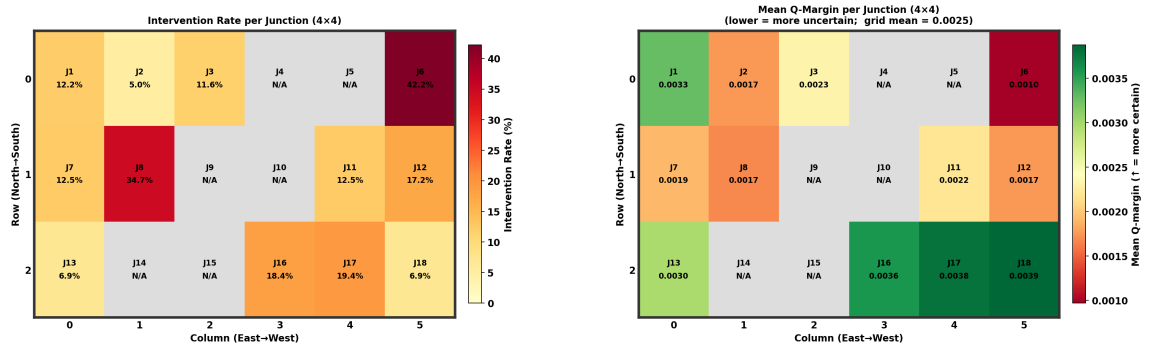


Figure 6: Per-junction intervention characteristics for the 4×4 grid. Top: intervention, override, anomaly trigger, and safety adjustment rates. Bottom: mean Q-margin at LLM invocation ($\tau = 0.05$, grid mean = 0.0025).

42.2% and J8 at 34.7%) also exhibit the lowest Q-margins (0.0010 and 0.0017 respectively), confirming that spatial intervention frequency is driven by genuine value uncertainty rather than arbitrary scheduling.

The 7×28 heatmap gives a network-scale interpretation. Intervention hotspots appear in the central arterial corridor, especially in columns 8–16 and rows 2–4, where rates approach 2.0%. These locations likely experience higher interaction pressure because they connect multiple traffic streams and receive congestion propagated from upstream and downstream directions. The corresponding Q-margin heatmap (grid mean = 0.0067) shows markedly lower confidence in these central regions, with values approaching 0.000, while peripheral junctions maintain higher margins above 0.006. This spatial alignment between intervention frequency and value uncertainty confirms that SafeGAT does not merely call the LLM more often in a larger network; it concentrates calls where the traffic network is structurally and dynamically more difficult to control.

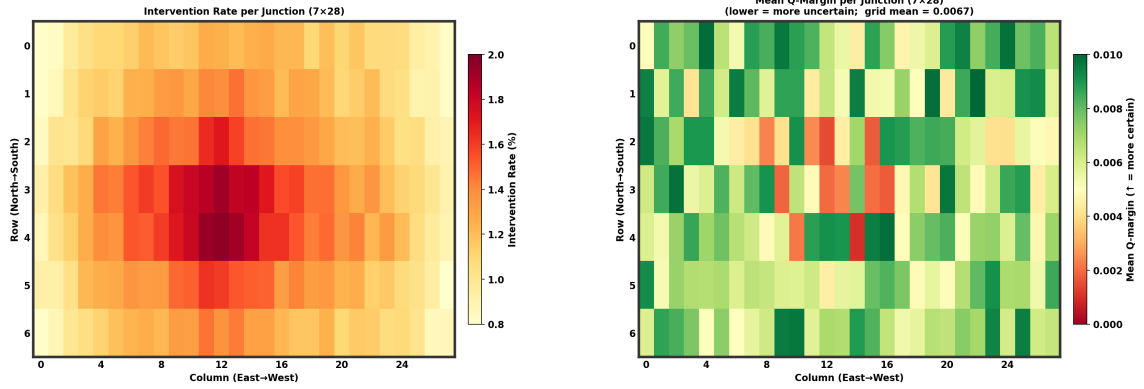


Figure 7: Per-junction intervention characteristics for the 7x28 grid. Left: intervention rate heatmap showing hotspots in columns 8–16 and rows 2–5. Right: mean Q-margin heatmap, where lower values indicate higher decision uncertainty.

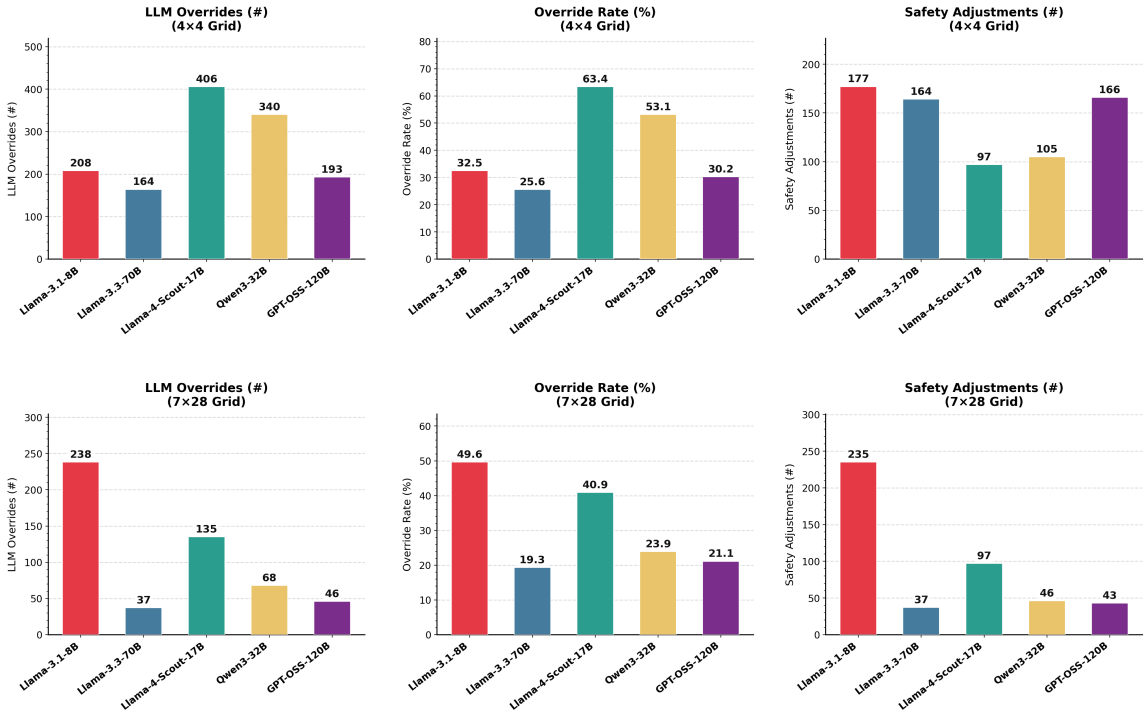


Figure 8: LLM intervention behaviour across network scales. Top (4x4): overrides, override rate, and safety adjustments per backend. Bottom (7x28): same metrics at larger scale.

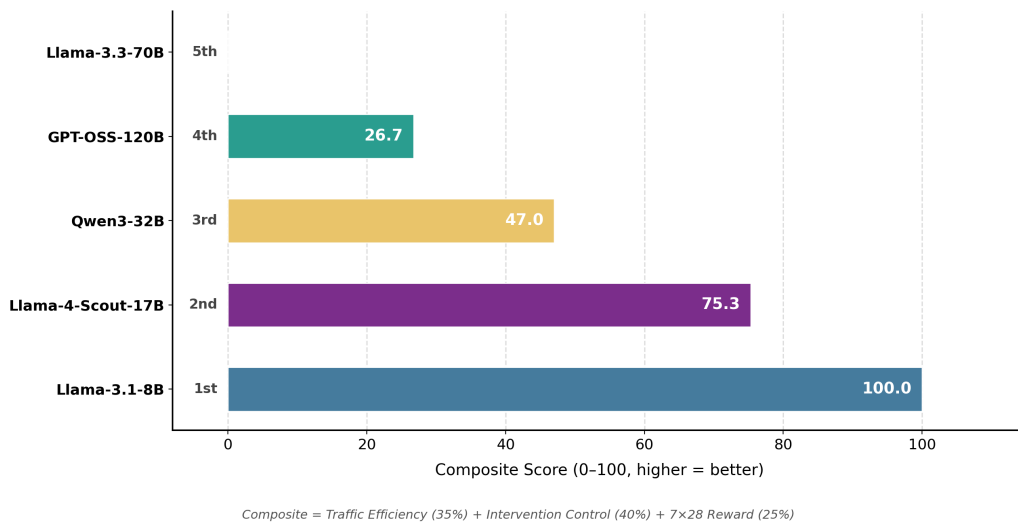


Figure 9: Overall LLM ranking by composite score (0–100). Composite = Traffic Efficiency (35%) + Intervention Control (40%) + 7x28 Reward (25%). The ranking should be interpreted as backend compatibility with the SafeGAT supervision structure, not as a universal ranking of model capability.

7.7. LLM Backend Comparison

Figures 8–10 compare the behavior of different LLM backends under the same SafeGAT framework. This experiment addresses whether larger LLMs automatically produce better traffic-control behavior. The results show that they do not. Although all backends receive the same structured control context, their accept/override patterns differ substantially. Llama-4-Scout-17B is the most aggressive backend on the 4×4 grid, with a 63.4% override rate. Llama-3.3-70B is more conservative, with a 25.6% override rate on the 4×4 grid and 19.3% on the 7×28 grid. This variation matters because SafeGAT requires calibrated advisory behavior, not maximum intervention. An overly aggressive LLM may override stable RL decisions and create more safety-shield corrections. A highly conservative LLM may preserve RL behavior but fail to correct genuinely risky states. Therefore, the best backend is the one that balances responsiveness with restraint. Llama-3.1-8B performs strongly because its override rate increases from 32.5% on the 4×4 grid to 49.6% on the 7×28 grid, matching the increased complexity and uncertainty of the larger network.

The trigger statistics explain this behavior further. For Llama-3.1-8B on the 4×4 grid, `anomaly_detected` and `low_confidence` each account for 638 triggers. This balance indicates that the backend responds to both traffic-state abnormality and value-function uncertainty. A backend dominated by only one trigger type would be less reliable because it might overreact to anomalies or ignore uncertainty in Q-values. The composite ranking therefore measures compatibility with the SafeGAT supervision structure rather than general model capability. The result suggests that model size is less important than calibrated accept/override behavior under structured risk signals.

7.8. Latency and Deployment Feasibility

Figure 11 evaluates whether SafeGAT can satisfy real-time execution constraints. This issue is critical because LLM-assisted control can be impractical if every intersection requires a model call at every control step. In a large traffic network, uniform LLM invocation would cause inference cost to scale with the number of intersections. SafeGAT avoids this by restricting intervention through the Top- K gate. Therefore, latency depends on the number of selected high-risk intersections rather than the total number of intersections. The latency distribution shows mean LLM latency of 250 ms, p95 latency of 600 ms, and p99 latency of 1200 ms under a 1000 ms step budget. These values indicate that single-call latency is usually acceptable, but synchronous multi-call execution can exceed the control budget. The synchronous-versus-asynchronous comparison confirms this: synchronous execution crosses the 1000 ms limit at around 1.5 LLM calls per step, while asynchronous execution remains close to 250 ms overhead even at 2.0 calls per step. This demonstrates that deployment feasibility depends on both selective gating and asynchronous execution.

The viability heatmaps show that SafeGAT remains feasible under asynchronous execution across the observed intervention range. This is a direct consequence of the framework design. The gate reduces the number of calls, the Top- K budget caps worst-case reasoning demand, fallback behavior handles late or invalid LLM responses, and the safety projection ensures that the controller can always execute a valid action. Therefore, real-time feasibility is not obtained by assuming the LLM is always fast; it is obtained by designing the control architecture so that LLM delay cannot block safe execution.

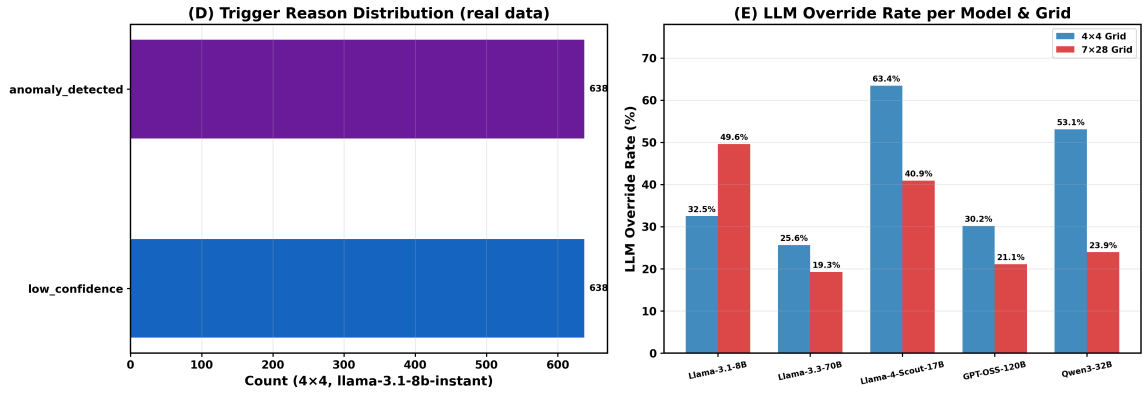


Figure 10: LLM trigger and decision statistics. Left: trigger reason distribution for 4x4 using Llama-3.1-8B, where `anomaly_detected` and `low_confidence` each account for 638 triggers. Right: override rate per model and grid, showing that Llama-3.1-8B scales from 32.5% to 49.6% as network complexity increases.

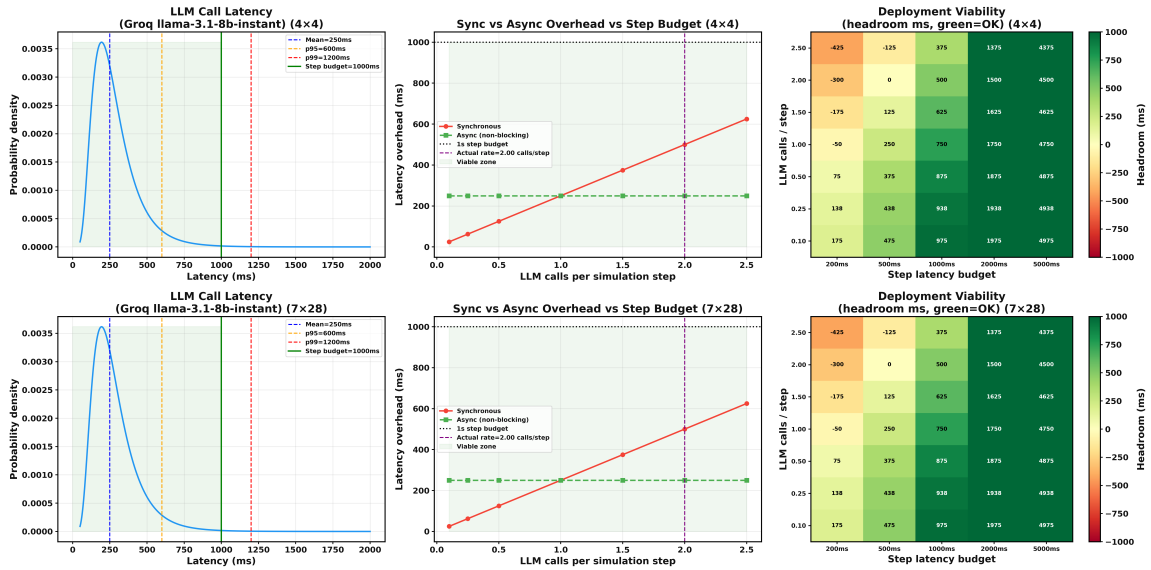


Figure 11: Latency analysis for 4x4 (top) and 7x28 (bottom). Left: LLM call latency distribution with mean 250 ms, p95 600 ms, p99 1200 ms, and a 1000 ms step budget. Centre: synchronous versus asynchronous overhead. Right: deployment viability heatmap, where green indicates feasible operation.

7.9. Discussion

The results support the central design claim of SafeGAT: LLM reasoning improves traffic signal control only when it is invoked selectively, bounded computationally, and constrained before execution. Across the 4x4 and 7x28 networks, SafeGAT improves the comparable traffic-efficiency metrics by reducing average travel time, queue length, and delay while increasing throughput. The gains over the graph-based RL backbone are smaller than the gains over classical control, but they are methodologically important because they show that uncertainty-gated LLM supervision improves an already strong spatial RL controller. The convergence and inference results further indicate that selective LLM supervision does not destabilize the GAT-DQN policy because the LLM is not used as a continuously active controller. The ablation study confirms this point: uniform LLM intervention increases calls and can degrade behavior, whereas risk-gated intervention combined with safety projection reduces LLM calls and eliminates violations under the implemented legal-phase and minimum-green constraints.

The robustness, spatial, backend, and latency analyses explain why the supervisory design is necessary. Under disturbance, SafeGAT becomes more useful because uncertainty and anomaly signals increase, activating additional

reasoning at states where value estimates are less reliable. Spatial heatmaps show that interventions concentrate in structurally difficult and low-confidence regions, especially in the central arterial corridor of the 7×28 network, rather than being uniformly distributed. The LLM backend comparison shows that model size alone does not determine control quality; what matters is calibrated accept/override behavior under the supplied risk context. Finally, the latency results show that real-time feasibility depends on architectural control over LLM usage. Top- K gating, asynchronous execution, fallback handling, and safety projection prevent LLM delay from blocking safe phase execution. Overall, SafeGAT converts LLM integration from heuristic action augmentation into a principled supervisory mechanism in which the RL policy proposes, the uncertainty/anomaly gate selects when reasoning is justified, the LLM advises under bounded risk, and the safety projection layer determines the final executable action.

8. Conclusion

This paper presented **SafeGAT**, a risk-bounded supervisory framework for integrating LLM-based reasoning with graph-attention reinforcement learning for traffic signal control. Unlike direct or uniform LLM-control paradigms, SafeGAT preserves the GAT-DQN policy as the primary decision maker and restricts language-model intervention to states with measurable execution risk. Decision uncertainty is quantified through Q-value confidence margins, abnormal traffic conditions are captured through four explicitly specified anomaly detectors, which exponentially-smoothed queue-spike z-scores, hard-range sensor-corruption checks, static capacity-threshold spillback flags, and simulator-provided emergency event signals and a Top- K intervention budget limits LLM refinement to the most critical intersections. All RL-proposed and LLM-refined actions are subsequently passed through a safety projection layer that enforces legal phase membership, valid phase transitions, and minimum green-time constraints before execution. This establishes a clear control hierarchy: the RL policy proposes, the uncertainty/anomaly gate decides whether external reasoning is justified, the LLM acts as a bounded advisor, and the safety layer determines the final executable phase.

Experiments on SUMO 4×4 and 7×28 networks demonstrate that SafeGAT improves traffic efficiency, robustness, and modeled safety compliance across both moderate and large-scale settings. SafeGAT achieves lower average travel time, queue length, and delay while improving throughput compared with classical control, adaptive control, RL-only baselines, graph-based RL, and LLM-assisted methods. The ablation study shows that uniform LLM intervention is inefficient and can degrade control quality, whereas risk-gated intervention reduces LLM calls by approximately 4× while maintaining stronger performance and eliminating modeled safety violations. Spatial analysis confirms that interventions concentrate in structurally difficult and low-confidence regions rather than being distributed uniformly, and robustness experiments show stronger recovery under disturbance because additional reasoning is activated when uncertainty or anomaly signals increase. LLM backend analysis further indicates that calibrated accept/override behavior is more important than model size, while latency results show that Top- K gating, asynchronous execution, fallback handling, and safety projection make the framework feasible within practical control budgets.

Limitations. Several fundamental constraints bound the current scope of SafeGAT. First, all experiments are conducted in microscopic simulation using SUMO; transitioning to physical deployment involves sensor noise characteristics, communication latency profiles, actuator response curves, and intersection geometry variability that no current simulator faithfully captures at scale, requiring standardized real-world testbeds that do not yet exist for systematic RL-based TSC evaluation. Second, the anomaly detectors rely on hand-specified thresholds because no sufficiently general learned anomaly detector currently transfers reliably across heterogeneous urban environments without per-city recalibration, as queue behavior, occupancy, and demand distributions vary across cities and infrastructure types in ways that make universal threshold learning an open problem. Third, the safety projection layer enforces only formally encodable feasibility constraints since the broader space of real urban safety requirements, including pedestrian signal phases, protected-permissive left-turn conflicts, and multimodal priority arbitration, currently lacks a unified formal representation amenable to deterministic projection. Fourth, the LLM is queried with a fixed prompt structure because no established methodology exists for systematically optimizing prompt design under closed-loop traffic control feedback, and prompt sensitivity remains poorly understood in structured decision support contexts. Fifth, the intervention budget K and risk-score coefficients are fixed by offline grid search because online adaptive parameter tuning under non-stationary traffic distributions remains computationally intractable at real-time signal management frequencies. Sixth, the framework operates on single-step phase decisions because integrating cycle-level

arterial coordination into a reinforcement learning framework without exponential growth in the joint action space is a combinatorially hard problem that current MARL methods cannot tractably solve at network scale.

Future Work. Long-term extensions of SafeGAT require foundational advances beyond the current state of the field. The simulation-to-reality gap will only be meaningfully closed once city-scale, high-fidelity trajectory datasets with full lane-level resolution, sensor fault annotations, and ground-truth phase records become publicly available, representing a decade-scale infrastructure challenge rather than an algorithmic one. On anomaly detection, a generalizable learned gate would require a universal model of normal traffic behavior across heterogeneous intersection types and demand regimes, demanding large-scale multi-city data harmonization efforts not currently underway in any coordinated form. On safety constraints, formally verifying the full operational safety envelope of signalized intersections would require advances in formal methods for hybrid dynamical systems capable of handling the stochasticity and partial observability of real traffic, a class of problems that remains open in control theory. On prompt robustness, understanding LLM override quality under prompt variation requires interpretability tools capable of attributing model decisions to specific input components under distributional shift, tools that do not yet exist for safety-critical control loops. On planning horizon, enabling the LLM to reason over multi-step phase sequences in real time would require either substantially faster inference than current language models provide or a theoretical framework integrating symbolic temporal reasoning with neural value estimation that does not yet exist. Finally, providing formal end-to-end safety guarantees for the full SafeGAT pipeline would require advances in neural network verification and probabilistic contract-based design at the scale and stochasticity of large urban traffic networks, a frontier that remains beyond current verification technology.

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A. LLM Prompt Template

This appendix provides the complete prompt template used by SafeGAT when querying the LLM advisor for intersection $i \in \mathcal{I}_t$ at control step t . The template is a plain-text string; all fields enclosed in braces are substituted with numerical or categorical values before the prompt is submitted to the model. The six information categories described in Section 4.4 correspond to the six blocks of the template below.

SafeGAT LLM Prompt Template

```

You are a traffic-signal safety verifier for intersection {junction_id}.
Your task is NOT to control all intersections but ONLY to decide whether
the RL agent's proposed action should be accepted or overridden for this
one intersection. You must return a JSON object and nothing else.

=== (i) Context ===
Simulation step      : {sim_step}
Current phase       : {current_phase}
Legal phases        : {legal_phases} [list of valid phase indices]

=== (ii) Lane-Level Traffic State ===
Queue lengths       : {queue} [vehicles per lane, list]
Occupancy           : {occ} [fraction 0-1 per lane, list]
Waiting time        : {wait} [cumulative seconds per lane, list]

=== (iii) Neighbourhood Summary ===
Neighbour occ.      : {nb_occ} [dict: junctionID -> mean occupancy]

=== (iv) RL Agent Proposal ===
Proposed phase      : {rl_phase} [integer index into legal_phases]
Q-value vector      : {q_values} [list, index matches legal_phases]
Confidence margin   : {margin} [normalised gap between top-2 Q-values]

=== (v) Risk Signals ===
Anomaly tags        : {anomaly_tags} [subset of: queue_spike,
sensor_corrupt, spillback, emergency]
Anomaly score       : {anomaly_score} [integer 0-4]

=== (vi) Intervention Severity ===
Risk score          : {severity} [float, higher = more critical]

=== Instructions ===
1. If the RL proposal appears appropriate given the above context, return
decision "accept" and set final_phase to the proposed phase.
2. If you believe a different phase from legal_phases would better handle
the current risk, return decision "override" and set final_phase to
that phase index.
3. final_phase MUST be an integer from the legal_phases list above.
4. reason must be a single concise sentence (no newlines).

Respond ONLY with a valid JSON object (no markdown, no preamble):
{"decision": "accept" | "override",
 "final_phase": <int>,
 "reason": "<one concise sentence>"}

```

Field substitution details. Table 3 summarises each template field, its data type, and where it originates in the SafeGAT pipeline. All numerical vectors are formatted as Python-style lists of floating-point values rounded to three decimal places. The `anomaly_tags` field is a list of strings drawn from the four anomaly categories defined in Section 4.2; an empty list is supplied when no anomaly is active. The `legal_phases` field lists the integer indices of all currently reachable phases from $\mathcal{A}_i^{legal}$; the ordering of this list is fixed at network initialization and matches the ordering of the `q_values` list, so that index k in both lists always refers to the same physical phase. This bijection is what allows the `final_phase` integer returned by the LLM to be mapped directly to a discrete signal-control action without any additional lookup.

Output parsing and phase-index mapping. The raw string returned by the LLM is processed by a two-stage parser. In stage one, the parser attempts `json.loads()` on the entire response string. In stage two, if stage one raises a `JSONDecodeError` (e.g., due to a markdown fence, a leading preamble sentence, or trailing commentary), the parser applies the following regex patterns to extract the required fields:

Table 3

Prompt template field reference. Each field is substituted before the prompt is submitted to the LLM. The `legal_phases` and `q_values` lists share the same index ordering so that `final_phase` maps unambiguously to a phase action.

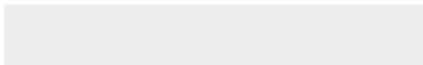
Field	Type	Source in SafeGAT
<code>junction_id</code>	string	Intersection identifier in SUMO network
<code>sim_step</code>	int	Current simulation time step t
<code>current_phase</code>	int	Active phase index p_i^t
<code>legal_phases</code>	int list	$\mathcal{A}_i^{legal}$, fixed ordering
<code>queue</code>	float list	Per-lane q_i^t from SUMO detector
<code>occ</code>	float list	Per-lane occupancy $o_i^t \in [0, 1]$
<code>wait</code>	float list	Per-lane cumulative waiting time w_i^t
<code>nb_occ</code>	dict	Mean occupancy per neighbor in $\mathcal{N}(i)$
<code>rl_phase</code>	int	$a_{RL,i}^t$ from GAT-DQN policy
<code>q_values</code>	float list	Q_i^t vector, same ordering as <code>legal_phases</code>
<code>margin</code>	float	Normalized confidence margin c_i^t (Eq. (24))
<code>anomaly_tags</code>	str list	Active anomaly labels from $\{A_{q,i}^t, A_{corrupt,i}^t, A_{spill,i}^t, A_{emer,i}^t\}$
<code>anomaly_score</code>	int	Aggregate $A_i^t \in \{0, 1, 2, 3, 4\}$ (Eq. (30))
<code>severity</code>	float	Risk score ρ_i^t (Eq. (31))

- Decision: "decision"\s*:\s*"accept|override)"
- Phase: "final_phase"\s*:\s*"([0-9]+)"

A response is accepted as valid and applied only when (1) both fields are successfully extracted, (2) the decision is one of `accept` or `override`, and (3) the extracted `final_phase` integer belongs to `legal_phases`. If any condition is violated, or if the call times out (500 ms wall-clock budget), the system reverts to the RL-proposed action $a_{RL,i}^t$ without modification. The `final_phase` integer is mapped to a signal-control action by looking up its position in the `legal_phases` list, which is identical in ordering to the Q-value vector Q_i^t ; this mapping is therefore a direct array lookup with $O(|\mathcal{A}_i^{legal}|)$ cost and requires no additional reasoning or disambiguation.

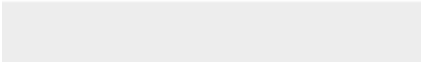


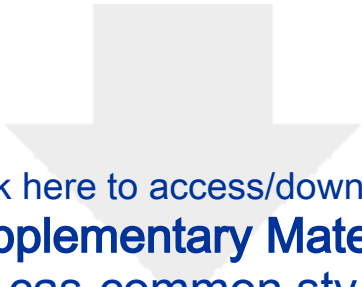
Click here to access/download
Supplementary Material
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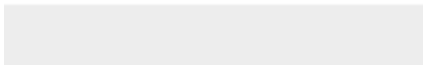
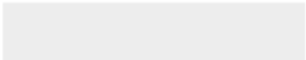


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Highlights

SafeGAT: Uncertainty-Gated and Safety-Constrained LLM Supervision for Graph Attention Reinforcement Learning in Traffic Signal Control

- SafeGAT: uncertainty-gated LLM supervision for graph-attention RL in TSC
- Q-margin and anomaly signals gate LLM calls to high-risk intersections only
- Top-K budget caps LLM overhead at $O(KC_{LLM})$ regardless of network size
- Deterministic safety layer enforces legal phases and minimum green times
- Reduces travel time, queue length, and delay; eliminates safety violations

Declaration of interests

☒The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

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